Chapter - 1

Calculus of Two and More Variables

1.1 Partial Differentiation

In many engineering scenarios, variables often depend on two or more independent variables. To illustrate, consider the example of a triode valve in electronic engineering, where the electron flow from the cathode C is regulated by the independent voltage v_g and v_p applied to the grid and plate of the valve, respectively. Figure 1.1 depicts how the plate current, i_p, is influenced by two independent variables, v_p and v_g. Understanding the variation of i_p necessitates familiarity with partial differentiation techniques. Numerous engineering challenges demand a clear grasp of these methods.

A function involving multiple variables, known as a function of several variables, requires the application of differential calculus. This branch often referred to as calculus of several variables, primarily focuses on functions of two or three variables.

Consider familiar examples such as the area of a rectangle, given by

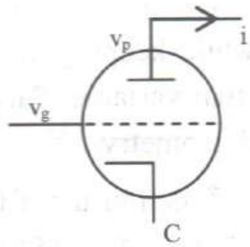


Figure 1.1 A function of two variables.

$$A = x\,y,$$

where A has a specific value for any pair of x and y values. Similarly, the volume of a parallelepiped, depending on three variables - length (x), breadth (y), and height (h) is expressed as

$$v = x\,y\,h.$$

If z represents a function of two independent variables x and y, then it is denoted by $z = f(x, y)$ or $\varphi(x, y)$. Likewise, if u represents a function of three variables x, y, and z, expressed as $u = f(x, y, z)$. In geometric terms, (x, y) denote the coordinates of a point in the xy-plane, with z representing the height of the surface $z = f(x, y)$. A set of points lying within a circle centered at (a, b) with a radius δ is considered a neighborhood of (a, b) within the circular region

$$R: (x - a)^2 + (y - b)^2 < \delta^2, \quad \text{or} \quad \{(x, y): |x - a| < \delta, |y - b| < \delta\}.$$

When z depends on three or more variables $(x, y, t, ...)$, we represent the relation as $z = f(x, y, t, ...)$.

Limit and continuity

Let $z = f(x, y)$ be a function of two independent variables x and y defined in the neighborhood of (a, b). Then the function f(x, y) is said to have **limit l** as (x, y) tends to (a, b) if and only if, for any positive number ε, there exists another positive number δ such that for every (x, y) in R such that

$$(x - a)^2 + (y - b)^2 < \delta^2 \text{ it implies } |f(x, y) - l| < \varepsilon.$$

We denote this as $\lim\limits_{(x,y)\to(a,b)} f(x, y) = l$

A function f(x, y) is said to be *continuous* at the point (a, b) if the limit

$$\lim_{(x, y) \to (a, b)} f(x, y) \text{ exists, and } \lim_{(x, y) \to (a, b)} f(x, y) = f(a, b).$$

If the function is continuous at all points of the neighborhood of (a, b), then it is said to be continuous in that region. A function that is not continuous at a point is said to be discontinuous at that point.

1.1.1 Partial Derivatives of First and Higher Order

Partial differentiation, a fundamental concept in calculus, is employed to determine how a function changes when one variable varies while all others are held constant. In multivariable calculus, functions often rely on more than one variable. By computing partial derivatives, we can analyze how the function responds exclusively to changes in a single variable, while the others are fixed. This process is denoted by symbols like ∂ (the partial derivative symbol) and ∂x (indicating the partial derivative with respect to x). This tool is crucial in various fields such as physics, economics, and engineering, where understanding how changes in one variable influence the entire system is paramount. The ordinary derivative of a function of multiple variables concerning one independent variable, while keeping all others constant, is termed the partial derivative of the function with respect to that variable. Partial derivatives find applications in disciplines like vector calculus and differential geometry.

Consider a function u = f (x, y) of two independent variables x and y. The derivative of u with respect x, treating y as constant, is called the *partial derivative of u with respect to x*, denoted by one of the symbols:

$$\frac{\partial u}{\partial x}, \ \frac{\partial f}{\partial x}, \ f_x(x, y), \ f_x, \ u_x.$$

This defined as $\frac{\partial u}{\partial x} = \lim_{\Delta x \to 0} \frac{f(x + \Delta x, y) - f(x, y)}{\Delta x}$ provided the limit exist.

Similarly, the partial derivative of u with respect to y, treating x as constant, is denoted by:

$$\frac{\partial u}{\partial y}, \ \frac{\partial f}{\partial y}, \ f_y(x, y), \ u_y, \ f_y$$

defined as:

$$\frac{\partial u}{\partial y} = \lim_{\Delta y \to 0} \frac{f(x, y + \Delta y) - f(x, y)}{\Delta y} \text{ provided the limit exists.}$$

The partial derivatives at a particular point (a, b) are often defined as:

$$\left(\frac{\partial u}{\partial x}\right)_{\text{at } (a, b)} \text{ or } f_x(a, b) = \lim_{h \to 0} \frac{f(a + h, b) - f(a, b)}{h}, \text{ and}$$

$$\left(\frac{\partial u}{\partial y}\right)_{\text{at } (a, b)} \text{ or } f_y(a, b) = \lim_{k \to 0} \frac{f(a, b + k) - f(a, b)}{k}$$

provided these limit exist.

Similarly, for a function u of three variables x, y, z, the partial derivatives with respect to x, y and z are defined as:

$$\frac{\partial u}{\partial x} = \lim_{\Delta x \to 0} \frac{f(x + \Delta x,\ y,\ z) - f(x,y,z)}{\Delta x \cdot}, \qquad \frac{\partial u}{\partial y} = \lim_{\Delta x \to 0} \frac{f(x,\ y + \Delta y,\ z) - f(x,\ y,\ z)}{\Delta y}$$

and $$\frac{\partial u}{\partial z} = \lim_{\Delta x \to 0} \frac{f(x,\ y, z + \Delta z) - f(x,y,z)}{\Delta z},$$ provided these limits exist.

In general, f_x and f_y are also functions of x and y, so they can be further differentiated partially with respect to x and y. Thus

$$\frac{\partial}{\partial x}\left(\frac{\partial u}{\partial x}\right) = \frac{\partial^2 u}{\partial x^2} \quad \text{or} \quad \frac{\partial^2 f}{\partial x^2} \quad \text{or} \quad \frac{\partial^2 u}{\partial x^2} \quad \text{or} \quad u_{xx} \quad \text{or} \quad f_{xx},$$

$$\frac{\partial}{\partial x}\left(\frac{\partial u}{\partial y}\right) = \frac{\partial^2 u}{\partial x^2} \quad \text{or} \quad \frac{\partial^2 f}{\partial x \partial y} \quad \text{or} \quad u_{xy} \quad \text{or} \quad f_{xy},$$

$$\frac{\partial}{\partial y}\left(\frac{\partial u}{\partial x}\right) = \frac{\partial^2 u}{\partial y \partial x} \quad \text{or} \quad \frac{\partial^2 f}{\partial y \partial x} \quad \text{or} \quad u_{yx} \quad \text{or} \quad f_{yx},$$

and $$\frac{\partial}{\partial y}\left(\frac{\partial u}{\partial y}\right) = \frac{\partial^2 u}{\partial y^2} \quad \text{or} \quad \frac{\partial^2 f}{\partial y^2} \quad \text{or} \quad u_{yy} \quad \text{or} \quad f_{yy}.$$

It can be verified that $\frac{\partial^2 u}{\partial x\, \partial y} = \frac{\partial^2 u}{\partial y\, \partial x}$, provided that the second order partial derivatives exist.

Worked Out Examples

1. Find the first and second partial derivatives of $x^3 + y^3 - 3axy$

Solution: Let $u = x^3 + y^3 - 3axy$

Partially differentiating with respect to *x* and *y*, respectively, we obtain

$$\frac{\partial u}{\partial x} = 3x^2 - 3ay, \quad \frac{\partial u}{\partial y} = 3y^2 - 3ax.$$

Additionally $\frac{\partial^2 u}{\partial x^2} = 6x, \quad \frac{\partial^2 u}{\partial y \partial x} = -3a, \quad \frac{\partial^2 u}{\partial y^2} = 6y, \quad \frac{\partial^2 u}{\partial x \partial y} = -3a.$

We observed that $\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 u}{\partial y \partial x}$.

2. If $u = x^2 + y^2 + z^2$, then show that $x\, u_x + y\, u_y + z\, u_z = 2u$

Solution: A function u of three variables x, y and z, is given by

$$u = x^2 + y^2 + z^2.$$

Partially differentiating it with respect to x, y and z respectively, we get

$$u_x = 2x, \quad u_y = 2y, \quad u_z = 2z.$$

We have $$x\, u_x + y\, u_y + z\, u_z = 2x^2 + 2y^2 + 2z^2 = 2\,(x^2 + y^2 + z^2) = 2u,$$

$\therefore$ $$x\, u_x + y\, u_y + z\, u_z = 2u.$$

3. If $V = \sqrt{x^2 + y^2 + z^2}$, then show that $V_{xx} + V_{yy} + V_{zz} = \frac{2}{V}$

Solution: A function V of three variables x, y, z, is given by

$$V = \sqrt{x^2 + y^2 + z^2} \quad \ldots(1)$$

Partially differentiating (1) with respect to *x*, we get

$$V_x = \frac{1}{2} \frac{2x}{\sqrt{x^2 + y^2 + z^2}} = \frac{x}{\sqrt{x^2 + y^2 + z^2}},$$

$$V_{xx} = \frac{1}{\sqrt{x^2 + y^2 + z^2}} + x\left(-\frac{1}{2}\right) \frac{1.\,2x}{(x^2 + y^2 + z^2)^{3/2}} = \frac{y^2 + z^2}{(x^2 + y^2 + z^2)^{3/2}}.$$

Partially differentiating (1) with respect to *y*, we get

$$V_y = \frac{1}{2} \frac{2y}{\sqrt{x^2 + y^2 + z^2}} = \frac{y}{\sqrt{x^2 + y^2 + z^2}},$$

$$V_{yy} = \frac{1}{\sqrt{x^2 + y^2 + z^2}} + y\left(-\frac{1}{2}\right) \frac{1.\,2y}{(x^2 + y^2 + z^2)^{3/2}} = \frac{z^2 + x^2}{(x^2 + y^2 + z^2)^{3/2}},$$

and partially differentiating (1) with respect to *z*, we get

$$V_z = \frac{1}{2} \frac{2z}{\sqrt{x^2 + y^2 + z^2}} = \frac{z}{\sqrt{x^2 + y^2 + z^2}},$$

$$V_{zz} = \frac{1}{\sqrt{x^2 + y^2 + z^2}} + z\left(-\frac{1}{2}\right) \frac{1.\,2z}{(x^2 + y^2 + z^2)^{3/2}} = \frac{x^2 + y^2}{(x^2 + y^2 + z^2)^{3/2}}.$$

We have

$$V_{xx} + V_{yy} + V_{zz} = \frac{y^2 + z^2}{(x^2 + y^2 + z^2)^{3/2}} + \frac{z^2 + x^2}{(x^2 + y^2 + z^2)^{3/2}} + \frac{x^2 + y^2}{(x^2 + y^2 + z^2)^{3/2}}$$

$$= \frac{y^2 + z^2 + z^2 + x^2 + x^2 + y^2}{(x^2 + y^2 + z^2)^{3/2}} = \frac{2(x^2 + y^2 + z^2)}{(x^2 + y^2 + z^2)^{3/2}}$$

$$= \frac{2}{\sqrt{x^2 + y^2 + z^2}} = \frac{2}{V}.$$

4. If $u = \log\sqrt{x^2 + y^2 + z^2}$, then show that $(x^2 + y^2 + z^2)\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}\right) = 1$

Solution: A function u of three variables x, y, z is given by

$$u = \log\sqrt{x^2 + y^2 + z^2}. \quad \ldots(1)$$

Partially differentiating (1) with respect to *x*, we get

$$\frac{\partial u}{\partial x} = \frac{1}{\sqrt{x^2 + y^2 + z^2}} \frac{1.\,2x}{2\sqrt{x^2 + y^2 + z^2}} = \frac{x}{x^2 + y^2 + z^2},$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{(x^2 + y^2 + z^2) - x.\,2x}{(x^2 + y^2 + z^2)^2} = \frac{y^2 + z^2 - x^2}{(x^2 + y^2 + z^2)^2}.$$

Partially differentiating (1) with respect to *y*, we get

$$\frac{\partial u}{\partial y} = \frac{1}{\sqrt{x^2+y^2+z^2}} \frac{1 . 2y}{2\sqrt{x^2+y^2+z^2}} = \frac{y}{x^2+y^2+z^2},$$

$$\frac{\partial^2 u}{\partial y^2} = \frac{(x^2+y^2+z^2) - y . 2y}{(x^2+y^2+z^2)^2} = \frac{z^2+x^2-y^2}{(x^2+y^2+z^2)^2}.$$

Partially differentiating (1) with respect to *z*, we get

$$\frac{\partial u}{\partial z} = \frac{1}{\sqrt{x^2+y^2+z^2}} \frac{1 . 2z}{2\sqrt{x^2+y^2+z^2}} = \frac{z}{x^2+y^2+z^2},$$

$$\frac{\partial^2 u}{\partial z^2} = \frac{(x^2+y^2+z^2) - z . 2z}{(x^2+y^2+z^2)^2} = \frac{x^2+y^2-z^2}{(x^2+y^2+z^2)^2}.$$

So
$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = \frac{y^2+z^2-x^2}{(x^2+y^2+z^2)^2} \frac{z^2+x^2-y^2}{(x^2+y^2+z^2)^2} + \frac{x^2+y^2-z^2}{(x^2+y^2+z^2)^2}$$

$$= \frac{y^2+z^2-x^2+ z^2+x^2-y^2+x^2 +y^2-x^2}{(x^2+y^2+z^2)^2} = \frac{1}{x^2+y^2+z^2}$$

$$\therefore \quad (x^2+y^2+z^2)\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}\right) = 1.$$

5. If $u = e^{xyz}$, then prove that $\frac{\partial^3 u}{\partial x\, \partial y\, \partial z} = (1 + 3xyz + x^2 y^2 z^2)\, e^{xyz}$.

Solution: Given $u = e^{xyz}$.

Partially differentiating it with respect to *x*, *y* and z respectively, we get

$$\frac{\partial u}{\partial z} = xy\, e^{xyz},$$

or
$$\frac{\partial^2 u}{\partial y \partial z} = e^{xyz}\, x + xy\, e^{xyz}\, xz = (x + x^2yz)\, e^{xyz},$$

or
$$\frac{\partial^3 u}{\partial x \partial y \partial z} = (x + x^2yz)\, e^{xyz}\, yz + e^{xyz}\,(1 + 2xyz),$$

$$\therefore \quad \frac{\partial^3 u}{\partial x \partial y \partial z} = e^{xyz}\,(1 + 3xyz + x^2y^2z^2).$$

6. If $z(x + y) = x^2 + y^2$, then show that $\left(\frac{\partial z}{\partial x} - \frac{\partial z}{\partial y}\right)^2 = 4\left(1 - \frac{\partial z}{\partial x} - \frac{\partial z}{\partial y}\right)$

Solution: A function z of two variables x, y, is given by

$$z\,(x + y) = x^2 + y^2. \qquad \ldots(1)$$

Partially differentiating (1) with respect to *x*, we get

$$z + (x + y)\frac{\partial z}{\partial x} = 2x, \qquad \therefore \quad \frac{\partial z}{\partial x} = \frac{2x - z}{x + y}.$$

Partially differentiating (1) with respect to *y*, we get

$$z + (x + y)\frac{\partial z}{\partial y} = 2y, \qquad \therefore \quad \frac{\partial z}{\partial y} = \frac{2y - z}{x + y}.$$

Now $$\left(\frac{\partial z}{\partial x}-\frac{\partial z}{\partial y}\right)^2=\left(\frac{2x-z}{x+y}-\frac{2y-z}{x+y}\right)^2=4\left(\frac{x-y}{x+y}\right)^2 \quad ...(2)$$

Also $$4\left(1-\frac{\partial z}{\partial x}-\frac{\partial z}{\partial y}\right)=4\left(1-\frac{2x-z}{x+y}-\frac{2y-z}{x+y}\right)=\frac{4\,(x+y-2x+z-2y+z)}{x+y}$$

$$=\frac{4\,(2z-x-y)}{x+y}=\frac{4}{(x+y)}\left[\frac{2(x^2+y^2)}{x+y}-\frac{(x+y)}{1}\right]$$

$$=\frac{4}{(x+y)}\,\frac{(2x^2+2y^2-x^2-2xy-y^2)}{(x+y)}=4\left(\frac{x-y}{x+y}\right)^2 \quad ...(3)$$

From (2) and (3), we get

$$\left(\frac{\partial z}{\partial x}-\frac{\partial z}{\partial y}\right)^2=4\left(1-\frac{\partial z}{\partial x}-\frac{\partial z}{\partial y}\right).$$

7. If $f=e^{x/y}+e^{y/z}+e^{z/x}$, then show that $x\frac{\partial f}{\partial x}+y\frac{\partial f}{\partial y}+z\frac{\partial f}{\partial z}=0$

Solution: A function f of three variables x, y, z, is given by

$$f(x, y, z)=e^{x/y}+e^{y/z}+e^{z/x}. \quad ...(1)$$

Partially differentiating (1) with respect to x considering y and z constant, we get

$$\frac{\partial f}{\partial x}=\frac{1}{y}e^{x/y}-\frac{z}{x^2}e^{z/x}.$$

Partially differentiating (1) with respect to y considering z and x constant, we get

$$\frac{\partial f}{\partial y}=-\frac{x}{y^2}e^{x/y}+\frac{1}{z}e^{y/z}.$$

Partially differentiating (1) with respect to z considering x and y constant, we get

$$\frac{\partial f}{\partial z}=-\frac{y}{z^2}e^{y/z}+\frac{1}{x}e^{z/x}.$$

Now $$x\frac{\partial f}{\partial x}+y\frac{\partial f}{\partial y}+z\frac{\partial f}{\partial z}=\frac{x}{y}e^{x/y}-\frac{z}{x}e^{z/x}-\frac{x}{y}e^{x/y}+\frac{y}{z}e^{y/z}-\frac{y}{z}e^{y/z}+\frac{z}{x}e^{z/x}=0.$$

8. If $u=\frac{z}{x}+\frac{y}{z}$, then show that $x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}+z\frac{\partial u}{\partial z}=0$

Solution: Given that $u=\frac{z}{x}+\frac{y}{z}$ ...(1)

Partially differentiating (1) with respect to x, y and z respectively, we get

$$\frac{\partial u}{\partial x}=-\frac{z}{x^2},\quad \frac{\partial u}{\partial y}=\frac{1}{z}\quad \text{and}\quad \frac{\partial z}{\partial x}=\frac{1}{x}-\frac{y}{z^2}.$$

Now $$x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}+z\frac{\partial u}{\partial z}=-\frac{z}{x}+\frac{y}{z}+\frac{z}{x}-\frac{y}{z}=0.$$

9. If $x=r\cos\theta$, $y=r\sin\theta$, prove that $\frac{\partial^2 r}{\partial x^2}+\frac{\partial^2 r}{\partial y^2}=\frac{1}{r}\left[\left(\frac{\partial r}{\partial x}\right)^2+\left(\frac{\partial r}{\partial y}\right)^2\right]$

Solution: The functions x and y of two variables r and θ are given by

$$x = r\cos\theta,\ y = r\sin\theta.$$

Squaring and adding, we get $x^2 + y^2 = r^2$. ...(1)

Partially differentiating (1) with respect to *x*, we get

$$2x = 2r\frac{\partial r}{\partial x}, \qquad \therefore \quad \frac{\partial r}{\partial x} = \frac{x}{r} = \frac{x}{\sqrt{x^2+y^2}}. \qquad ...(2)$$

Partially differentiating (1) with respect to *y*, we get

$$2y = 2r\frac{\partial r}{\partial y}, \qquad \therefore \quad \frac{\partial r}{\partial y} = \frac{y}{r} = \frac{y}{\sqrt{x^2+y^2}}. \qquad ...(3)$$

Partially differentiating with respect to *x* to (2), we get

$$\frac{\partial^2 r}{\partial x^2} = \frac{r - x\frac{\partial r}{\partial x}}{r^2} = \frac{r - x\cdot\frac{x}{r}}{r^2} = \frac{r^2 - x^2}{r^3} = \frac{x^2 + y^2 - x^2}{(x^2+y^2)^{3/2}} = \frac{y^2}{(x^2+y^2)^{3/2}},$$

Partially differentiating with respect to *y* to (3) respectively, we get

$$\frac{\partial^2 r}{\partial y^2} = \frac{r - y\frac{\partial r}{\partial y}}{r^2} = \frac{r - y\cdot\frac{y}{r}}{r^2} = \frac{r^2 - y^2}{r^3} = \frac{x^2 + y^2 - y^2}{(x^2+y^2)^{3/2}} = \frac{x^2}{(x^2+y^2)^{3/2}}.$$

We have

$$\frac{\partial^2 r}{\partial x^2} + \frac{\partial^2 r}{\partial y^2} = \frac{y^2}{(x^2+y^2)^{3/2}} + \frac{x^2}{(x^2+y^2)^{3/2}} = \frac{x^2+y^2}{(x^2+y^2)^{3/2}} = \frac{1}{\sqrt{x^2+y^2}} = \frac{1}{r},$$

and

$$\frac{1}{r}\left[\left(\frac{\partial r}{\partial x}\right)^2 + \left(\frac{\partial r}{\partial y}\right)^2\right] = \frac{1}{r}\left(\frac{x^2}{r^2} + \frac{y^2}{r^2}\right) = \frac{1}{r}\ \frac{x^2+y^2}{r^2} = \frac{1}{r}.$$

Exercise - 1

1. Evaluate f_x and f_y of the following function f(x, y)

   (i) $x^2y - x\sin(xy)$ (ii) $\log(x^2 + y^2)$

   (iii) $\tan^{-1}\frac{x^2+y^2}{x+y}$ (iv) $\frac{x^2-y^2}{x^2+y^2}$

   (v) $\sin^{-1}(ax + by)$

2. Find the partial derivative of second orders of the following functions f(x, y)

   (i) $x^3 + 3x^2y + 3xy^2 + y^3$ (ii) $e^{x^2 + xy + y^2}$

   (iii) $\log(x^2y + xy^2)$ (iv) $\tan^{-1}\frac{2xy}{x^2+y^2}$

3. (i) If $u = \sin^{-1}\frac{x}{y} + \tan^{-1}\frac{y}{x}$, then show that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 0$

   (ii) If $u = x^2y + y^2z + z^2x$, then show $u_x + u_y + u_z = (x + y + z)^2$

(iii) If $u = \log \frac{x^2 + y^2}{x + y}$, then prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 1$

(iv) If $u = \log(x^2 + y^2 + z^2)$, then show that $x\, u_{yz} = y\, u_{zx} = z\, u_{xy}$

4. If $u = \log(x^3 + y^3 + z^3 - 3xyz)$, then show that

(i) $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = \frac{3}{x + y + z}$

(ii) $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = -\frac{3}{(x + y + z)^2}$

(iii) $\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right)^2 u = -\frac{9}{(x + y + z)^2}$

5. (i) If $u = f\left(\frac{y}{x}\right)$, then show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 0$

(ii) If $V = z \tan^{-1} \frac{y}{x}$, then show that $\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} = 0$

(iii) If $u = \tan(y + ax) - (y - ax)^{3/2}$, then show that $\frac{\partial^2 u}{\partial x^2} = a^2 \frac{\partial^2 u}{\partial y^2}$

(iv) If $z = \phi(x + ay) + \psi(x - ay)$, then show that $a^2 \frac{\partial^2 z}{\partial x^2} = \frac{\partial^2 z}{\partial y^2}$

6. If $u = f(xyz)$, then show that $x^2 u_{xx} = y^2 u_{yy} = z^2 u_{zz}$

7. If $u = \frac{1}{\sqrt{x^2 + y^2 + z^2}}$, then prove that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0$

8. If $x = r \cos\theta$, $y = r \sin\theta$, then show that

(i) $\frac{\partial r}{\partial x} = \frac{\partial x}{\partial r}$ (ii) $\frac{1}{r} \frac{\partial x}{\partial \theta} = r \frac{\partial \theta}{\partial x}$ (iii) $\frac{\partial^2 \theta}{\partial x^2} + \frac{\partial^2 \theta}{\partial y^2} = 0$

9. If $u = e^{r\cos\theta} \cos(r \sin\theta)$, $v = e^{r\cos\theta} \sin(r\sin\theta)$, then prove $\frac{\partial u}{\partial r} = \frac{1}{r}\frac{\partial v}{\partial \theta}$ and $\frac{\partial v}{\partial r} = -\frac{1}{r}\frac{\partial u}{\partial \theta}$

10. If $u = f(r)$, where $r^2 = x^2 + y^2$, then prove that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f''(r) + \frac{1}{r} f'(r)$

Answers

1. (i) $2xy - xy\cos(xy) - \sin(xy)$, $x^2(1 - \cos xy)$ (ii) $\frac{2x}{x^2 + y^2}$, $\frac{2y}{x^2 + y^2}$

(iii) $\frac{x^2 + 2xy - y^2}{(x^2 + y^2)^2 + (x + y)^2}$, $\frac{y^2 + 2xy - x^2}{(x^2 + y^2)^2 + (x + y)^2}$ (iv) $\frac{4xy^2}{(x^2 + y^2)^2}$, $-\frac{4x^2y}{(x^2 + y^2)^2}$

(v) $\frac{a}{\sqrt{1 - (ax + by)^2}}$, $\frac{b}{\sqrt{1 - (ax + by)^2}}$

2. (i) $6(x + y)$, $6(x + y)$, $6(x + y)$, $6(x + y)$

(ii) $e^{x^2+xy+y^2}\{(2x+y)^2+2\}$, $e^{x^2+xy+y^2}((2x+y)(x+2y)+1\}$,

$e^{x^2+xy+y^2}[(2x+y)(x+2y)+1]$, $e^{x^2+xy+y^2}[(x+2y)^2+2]$

(iii) $-\frac{1}{x^2}-\frac{1}{(x+y)^2}, -\frac{1}{(x+y)^2}, -\frac{1}{(x+y)^2}, -\frac{1}{y^2}-\frac{1}{(x+y)^2}$

1.1.2 Homogenous Functions

An expressions in terms of two independent variables x and y in the form

$$a_0x^n + a_1x^{n-1}y + a_2x^{n-2}y^2 + \dots\dots\dots + a_ny^n$$

in which every term is of the n^{th} degree, is called a *homogenous function of degree n*. It can be written as

$$x^n\left[a_0 + a_1\left(\frac{y}{x}\right) + a_2\left(\frac{y}{x}\right)^2 + \dots\dots\dots\dots + a_n\left(\frac{y}{x}\right)^n\right].$$

Thus, if any function f(x, y) of two independent variables x and y can be expressed in the form $x^n\phi\left(\frac{y}{x}\right)$, then it is called a *homogeneous function of degree n in x and y.*

Example *Let* $f(x, y) = 2x^3 + 3x^2y + y^3$ *then* $f(x, y) = x^3\left[2 + 3\frac{y}{x} + \left(\frac{y}{x}\right)^3\right] = x^3\phi\left(\frac{y}{x}\right)$.

Also

$$f(tx, ty) = 2t^3x^3 + 3t^2x^2\,ty + t^3y^3 = t^3(2x^3 + 3x^2y + y^3) = t^3 f(x, y)$$

Hence, the function f of two variables is homogenous of degree three.

In general, a function $f(x, y, z \dots\dots\dots)$ is said to be homogenous function of degree n in $x, y, z \dots \dots$ if it can be expressed in the form $x^n\phi\left(\frac{y}{x}, \frac{z}{x}, \dots\right)$,

or $f(tx, ty, tz, \dots\dots) = t^n f(x, y, z, \dots\dots\dots)$ for every values of t.

Euler's Theorem for two and three variables

Theorem 1 *If* $u = f(x, y)$ *be a homogenous function of two variables x and y of degree n, then* $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = nu$

Proof: Given that u is a homogenous function of x and y of degree n, so that the function u can be written as

$$u = x^n f\left(\frac{y}{x}\right). \qquad \dots(1)$$

Partially differentiating (1) with respect to x, we get

$$\frac{\partial u}{\partial x} = nx^{n-1}\cdot f\left(\frac{y}{x}\right) + x^n f'\left(\frac{y}{x}\right)\left(-\frac{y}{x^2}\right) = nx^{n-1}f\left(\frac{y}{x}\right) - y\,x^{n-2}f'\left(\frac{y}{x}\right), \qquad \dots(2)$$

and

$$\frac{\partial u}{\partial y} = x^n f'\left(\frac{y}{x}\right)\frac{1}{x} = x^{n-1}f'\left(\frac{y}{x}\right). \qquad \dots(3)$$

Multiplying (1) by x and (2) by y, and adding (1) and (2), we get

$$x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}= n x^n f\left(\frac{y}{x}\right) - x^{n-1} y f'\left(\frac{y}{x}\right) + x^{n-1} y f'\left(\frac{y}{x}\right) = n x^n f\left(\frac{y}{x}\right) = nu.$$

$$\therefore \quad x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}= n u.$$

Total Derivatives and Differentials

Let $u = f(x, y)$ where $x = \phi(t)$ and $y = \psi(t)$ then u can be expressed as a function of t alone and the ordinary derivative $\frac{du}{dt}$ is called *total differential coefficient* of u with respect to t.

1. *Find $\frac{du}{dt}$ without actually substituting the value of x and y in f(x, y)*

A function u of two variables x, y, is given by

$$u = f(x, y). \qquad \ldots(1)$$

Suppose Δt be increment of t whereas Δx, Δy and Δu be increments of x, y and u respectively, then

$$u + \Delta u = f(x + \Delta x, y + \Delta y). \qquad \ldots(2)$$

Subtracting (1) from (2), we get

$$\Delta u = f(x + \Delta x, y + \Delta y) - f(x, y)$$

$$= f(x + \Delta x, y + \Delta y) - f(x, y + \Delta y) + f(x, y + \Delta y) - f(x, y).$$

$$\frac{\Delta u}{\Delta t} = \frac{f(x + \Delta x, y + \Delta y) - f(x, y + \Delta y)}{\Delta x}\,\frac{\Delta x}{\Delta t} + \frac{f(x, y + \Delta y) - f(x, y)}{\Delta y}\cdot\frac{\Delta y}{\Delta t}.$$

Taking limits as $\Delta t \to 0$, Δx and Δy both tends to zero, we have

$$\lim_{\Delta t\to 0}\frac{\Delta u}{\Delta t} = \lim_{\Delta x\to 0}\frac{f(x + \Delta x, y + \Delta y) - f(x, y + \Delta y)}{\Delta x}\,\frac{dx}{dt} + \lim_{\Delta y\to 0}\frac{f(x, y + \Delta y) - f(x, y)}{\Delta y},$$

$$\frac{dy}{dt} \text{ or } \frac{du}{dt} = \frac{\partial f(x, y + \Delta y)}{\partial x}\,\frac{dx}{dt} + \frac{\partial f(x, y)}{\partial y}\,\frac{dy}{dt}.$$

Suppose that $\partial f(x, y)$ to be continuous function of y, we get

$$\frac{du}{dt} = \frac{\partial f(x, y)}{\partial x}\,\frac{dx}{dt} + \frac{\partial f(x, y)}{\partial y}\,\frac{dy}{dt},$$

$$\therefore \quad \frac{du}{dt} = \frac{\partial u}{\partial x}\,\frac{dx}{dt} + \frac{\partial u}{\partial y}\,\frac{dy}{dt}. \qquad \ldots(3)$$

Cor. 1 In particular if $u = f(x, y)$ where y is a function x, then putting $t = x$ to (1), we obtain

$$\frac{du}{dx} = \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y}\,\frac{dy}{dx}. \qquad \ldots(4)$$

Cor. 2 If $u = f(x, y)$, then the total differential of u is defined as

$$du = \frac{\partial u}{\partial x}\, dx + \frac{\partial u}{\partial y}\, dy.$$

Cor. 3 If u = f(x, y, z) where x, y, z are all functions of a single variable t, then we can similarly prove that

$$\frac{du}{dt} = \frac{\partial u}{\partial x}\frac{dx}{dt} + \frac{\partial u}{\partial y}\frac{dy}{dt} + \frac{\partial u}{\partial z}\frac{dz}{dt}.$$

Differentiation of Composite and implicit functions

If u = f(x, y) = c be implicit function which defines as a differentiable function of *x*, then (2) becomes

$$0 = \frac{du}{dx} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y}\frac{dy}{dx},$$

$$\therefore \quad \frac{dy}{dx} = -\frac{f_x}{f_y}, \quad f_y \neq 0.$$

Increments

If u = f(x, y) be function of two variables x and y and let Δx and Δy be theincrements of x and y respectively, then the increment Δu of u = f(x, y) is

$$\Delta u = f(x + \Delta x, y + \Delta y) - f(x, y).$$

Example Obtain the change in f(x, y) if (x, y) = $3x^2$ - xy changes from (1, 2) to (1.01, 1.98)

A function u of two variables x, y, is given by

$$u = f(x, y) = 3x^2 - xy. \quad \ldots(1)$$

Let Δu, Δx, Δy be small increments of u, x, y respectively, we get

$$u + \Delta u = f(x + \Delta x, y + \Delta y) = 3(x + \Delta x)^2 - (x + \Delta x)(y + \Delta y). \quad \ldots(2)$$

Subtracting (1) from (2), we get

$$\Delta u = f(x + \Delta x, y + \Delta y) - f(x, y) = 3(x + \Delta x)^2 - (x + \Delta x)(y + \Delta y) - (3x^2 - xy),$$

$$\therefore \quad \Delta u = 6x\,\Delta x + 3(\Delta x)^2 - x\,\Delta y - y\,\Delta x - \Delta x\,\Delta y. \quad \ldots(3)$$

Since (x, y) changes from (1, 2) to (1.01, 1.98), so substituting x = 1, y = 1, Δx = .01, Δy = - 0.02 in (3), we get

$$\Delta u = 6(1)(0.01)+3(0.01)^2 -1(-0.02)-2(0.01) - (0.01)(-0.02) = 0.0605.$$

This follows that the following definitions.

Definition 1.1 *If u = f(x, y) and Δx and Δy be increments of x and y respectively, then the differentials dx and dy of the independent variables x and y are dx = Δx and dy = Δy.*

Definition 1.2 *If u = f(x, y) be a function of two variables, then the differential du of the independent variable u is defined by* $du = f_x\,dx + f_y\,dy$.

Partial derivatives of a function of two functions

Theorem 2 *If $z = f(x, y)$ where $x = \phi_1(u, v)$, $y = \phi_2(u, v)$ and u, v are independent variables, then*

$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial u} \quad and \quad \frac{\partial z}{\partial v} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial v} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial v}$$

Proof: A function z of two variables x, y, is given by

$$z = f(x, y). \qquad \ldots(1)$$

Differentiating (1), we get

$$dz = \frac{\partial z}{\partial x}\,dx + \frac{\partial z}{\partial y}\,dy. \qquad \ldots(2)$$

A function x of two variables u, v, is given by

$$x = \phi_1(u, v). \qquad \ldots(3)$$

Differentiating (3), we get

$$dx = \frac{\partial x}{\partial u}\,du + \frac{\partial x}{\partial v}\,dv.$$

A function y of two variables u, v, is given by

$$y = \phi_2(u, v). \qquad \ldots(4)$$

Differentiating (4), we get

$$dy = \frac{\partial y}{\partial u}\,du + \frac{\partial y}{\partial v}\,dv.$$

Substituting the values of dx and dy in (2), we get

$$dz = \frac{\partial z}{\partial x}\left(\frac{\partial x}{\partial u}du + \frac{\partial x}{\partial v}dv\right) + \frac{\partial z}{\partial y}\left(\frac{\partial y}{\partial u}du + \frac{\partial y}{\partial v}dv\right)$$

$$dz = \left(\frac{\partial z}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial u}\right)du + \left(\frac{\partial z}{\partial x}\frac{\partial x}{\partial v} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial v}\right)dv. \qquad \ldots(5)$$

Since *z* is a function of two variables u and v,

$$dz = \frac{\partial z}{\partial u}\,du + \frac{\partial z}{\partial v}\,dv. \qquad \ldots(6)$$

Comparing (5) with (6), we get

$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial u} \quad \text{and} \quad \frac{\partial z}{\partial v} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial v} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial v}.$$

Cor. It can also be generalized in the case of more than two variables.

If $X = f(x, y, z)$ where $x = \phi_1(u, v, w)$, $y = \phi_2(u, v, w)$, $z = \phi_3(u, v, w)$ and u, v, w are independent variables, then

$$\frac{\partial X}{\partial u} = \frac{\partial X}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial X}{\partial y}\frac{\partial y}{\partial u} + \frac{\partial X}{\partial z}\frac{\partial z}{\partial u}, \qquad \frac{\partial X}{\partial v} = \frac{\partial X}{\partial x}\frac{\partial x}{\partial v} + \frac{\partial X}{\partial y}\frac{\partial y}{\partial v} + \frac{\partial X}{\partial z}\frac{\partial z}{\partial v},$$

$$\frac{\partial X}{\partial w} = \frac{\partial X}{\partial x}\frac{\partial x}{\partial w} + \frac{\partial X}{\partial y}\frac{\partial y}{\partial w} + \frac{\partial X}{\partial z}\frac{\partial z}{\partial w}.$$

Euler's Theorem on homogeneous functions of three variables

Theorem 3 *If $f(x, y, z)$ be a homogeneous function in x, y, z of degree n having continuous partial derivatives then* $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = n u$

Proof: Given that $u = f(x, y, z)$ be homogeneous function with degree n, then it can be written as

$$u = x^n f\left(\frac{y}{x}, \frac{z}{x}\right) = x^n f(v, w), \quad \ldots(1)$$

where $v = \frac{y}{x}$, $w = \frac{z}{x}$. ...(2)

Partially differentiating (2) with respect to x, y and z respectively, we get

$$\frac{\partial v}{\partial x} = -\frac{y}{x^2}, \quad \frac{\partial v}{\partial y} = \frac{1}{x}, \quad \frac{\partial v}{\partial z} = 0, \quad \text{and}$$

$$\frac{\partial w}{\partial x} = -\frac{z}{x^2}, \quad \frac{\partial w}{\partial y} = 0, \quad \frac{\partial w}{\partial z} = \frac{1}{x}.$$

Partially differentiating (1) with respect to x, we get

$$\frac{\partial u}{\partial x} = n x^{n-1} f(v, w) + x^n\left(\frac{\partial f}{\partial v}\frac{\partial v}{\partial x} + \frac{\partial f}{\partial w}\frac{\partial w}{\partial x}\right) = n x^{n-1} f(v, w) + x^n\left(-\frac{y}{x^2}\frac{\partial f}{\partial v} - \frac{z}{x^2}\frac{\partial f}{\partial w}\right),$$

$$\frac{\partial u}{\partial x} = nx^{n-1} f(v, w) - x^{n-2}\left(y\frac{\partial f}{\partial v} + z\frac{\partial f}{\partial w}\right).$$

Partially differentiating (1) with respect to y, we get

$$\frac{\partial u}{\partial y} = x^n \frac{\partial f}{\partial v}\frac{\partial v}{\partial y} = x^{n-1}\frac{\partial f}{\partial v}.$$

Partially differentiating (1) with respect to z, we get

$$\frac{\partial u}{\partial z} = x^n \frac{\partial f}{\partial w}\frac{\partial w}{\partial z} = x^{n-1}\frac{\partial f}{\partial w}.$$

Now $$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = nx^n f(v, w) - x^{n-1}\left(y\frac{\partial f}{\partial v} + z\frac{\partial f}{\partial w}\right) + x^{n-1} y\frac{\partial f}{\partial v} + z x^{n-1}\frac{\partial f}{\partial w}$$

$$= n x^n f\left(\frac{y}{x}, \frac{z}{x}\right) = n u$$

$$\therefore \quad x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = n u.$$

Note In general, if u is a function of degree in x, y, z ... ..., then

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} + \ldots\ldots\ldots = nu.$$

1.1.4 Jacobians and Their Properties

Jacobi (1804-1851), one of the most distinguished German scientists of the nineteenth century, advanced the theory of determinants and turned it into a crucial tool for assessing multiple integrals and differential equations. He also utilized transformation methods to analyze complex integrals, including those involved in calculating arc length. Similar to Euler, Jacobi was an extremely productive writer and an even more prolific calculator, contributing significantly to various applied mathematical fields.

Let u = u (x, y) and v = v (x, y) be functions of two variables x and y. The determinant defined as

$$\begin{vmatrix} \dfrac{\partial u}{\partial x} & \dfrac{\partial u}{\partial y} \\[2ex] \dfrac{\partial v}{\partial x} & \dfrac{\partial v}{\partial y} \end{vmatrix}$$

is called the *Jacobian* or the *Functional Determinant*, of the functions u and v with respect to x and y, and is denoted by

$$J\left(\frac{u, v}{x, y}\right) = \frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} \dfrac{\partial u}{\partial x} & \dfrac{\partial u}{\partial y} \\[2ex] \dfrac{\partial v}{\partial x} & \dfrac{\partial v}{\partial y} \end{vmatrix}.$$

Similarly, let u = u(x, y, z), v = v(x, y, z), and w = w(x, y, z) be functions of three variables x, y, and z. The determinant defined as

$$\begin{vmatrix} \dfrac{\partial u}{\partial x} & \dfrac{\partial u}{\partial y} & \dfrac{\partial u}{\partial z} \\[2ex] \dfrac{\partial v}{\partial x} & \dfrac{\partial v}{\partial y} & \dfrac{\partial v}{\partial z} \\[2ex] \dfrac{\partial w}{\partial x} & \dfrac{\partial w}{\partial y} & \dfrac{\partial w}{\partial z} \end{vmatrix},$$

is called the *Jacobian* or the *Functional Determinant* of the functions u, v and w with respect to x, y and z, and is denoted by

$$J\left(\frac{u, v, w}{x, y, z}\right) = \frac{\partial(u, v, w)}{\partial(x, y, z)} = \begin{vmatrix} \dfrac{\partial u}{\partial x} & \dfrac{\partial u}{\partial y} & \dfrac{\partial u}{\partial z} \\[2ex] \dfrac{\partial v}{\partial x} & \dfrac{\partial v}{\partial y} & \dfrac{\partial v}{\partial z} \\[2ex] \dfrac{\partial w}{\partial x} & \dfrac{\partial w}{\partial y} & \dfrac{\partial w}{\partial z} \end{vmatrix}.$$

In general, if $u_1 = u_1(x_1, x_2, \ldots x_n)$, $u_2 = u_2(x_1, x_2, \ldots x_n)$, …, and $u_n = u_n(x_1, x_2, \ldots x_n)$, then the *Jacobian* or the *Functional Determinant* of the functions $u_1, u_2, u_3, \ldots, u_n$ with respect to $x_1, x_2, \ldots, x_n$ is defined as

$$J\left(\frac{u_1, u_2, \ldots, u_n}{x_1, x_2, \ldots, x_n}\right) = \frac{\partial(u_1, u_2, \ldots, u_n)}{\partial(x_1, x_2, \ldots, x_n)} = \begin{vmatrix} \frac{\partial u_1}{\partial x_1} & \frac{\partial u_1}{\partial x_2} & \cdots & \frac{\partial u_1}{\partial x_n} \\ \frac{\partial u_2}{\partial x_1} & \frac{\partial u_2}{\partial x_2} & \cdots & \frac{\partial u_2}{\partial x_n} \\ \vdots & \vdots & \cdots & \vdots \\ \frac{\partial u_n}{\partial x_1} & \frac{\partial u_n}{\partial x_2} & \cdots & \frac{\partial u_n}{\partial x_n} \end{vmatrix}.$$

Property 1. *If $u = u(x, y)$ and $v = v(x, y)$ be the functions of two variables, then* $\frac{\partial(u, v)}{\partial(x, y)} \times \frac{\partial(x, y)}{\partial(u, v)} = 1$.

Proof: Here, $u = u(x, y)$ and $v = v(x, y)$, then

$$\frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix},$$

and the Jacobian of x, y with respect to u and v is defined as

$$\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}.$$

We have $$\frac{\partial(u, v)}{\partial(x, y)} \times \frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}$$

By changing the rows into column in second determinant, we get

$$= \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial y}{\partial u} \\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} \frac{\partial u}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial u}{\partial y}\frac{\partial y}{\partial u} & \frac{\partial u}{\partial x}\frac{\partial x}{\partial v} + \frac{\partial u}{\partial y}\frac{\partial y}{\partial v} \\ \frac{\partial v}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial v}{\partial y}\frac{\partial y}{\partial u} & \frac{\partial v}{\partial x}\frac{\partial x}{\partial v} + \frac{\partial v}{\partial y}\frac{\partial y}{\partial v} \end{vmatrix}$$

$$= \begin{vmatrix} \frac{\partial u}{\partial u} & \frac{\partial u}{\partial v} \\ \frac{\partial v}{\partial u} & \frac{\partial v}{\partial v} \end{vmatrix} = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1.$$

$$\frac{\partial(u, v)}{\partial(x, y)} \times \frac{\partial(u, v)}{\partial(x, y)} = 1.$$

Property 2. *If $u = u(r, s)$ and $v = v(r, s)$ are functions of two variables r and s, where $r = r(x, y)$, $s = s(x, y)$ be the functions of two variables x and y, then* $\dfrac{\partial(u, v)}{\partial(x, y)} = \dfrac{\partial(u, v)}{\partial(r, s)} \times \dfrac{\partial(r, s)}{\partial(x, y)}$.

Proof: Here, $u = u(r, s)$ and $v = v(r, s)$ are functions of two variables r and s, and $r = r(x, y)$, $s = s(x, y)$ are the function of two variables x and y, then the Jacobians are defined as

$$\frac{\partial(u, v)}{\partial(r, s)} = \begin{vmatrix} \frac{\partial u}{\partial r} & \frac{\partial u}{\partial s} \\ \frac{\partial v}{\partial r} & \frac{\partial v}{\partial s} \end{vmatrix}, \text{ and } \frac{\partial(r, s)}{\partial(x, y)} = \begin{vmatrix} \frac{\partial r}{\partial x} & \frac{\partial r}{\partial y} \\ \frac{\partial s}{\partial x} & \frac{\partial s}{\partial y} \end{vmatrix}.$$

We have

$$\frac{\partial(u, v)}{\partial(r, s)} \times \frac{\partial(r, s)}{\partial(x, y)} = \begin{vmatrix} \frac{\partial u}{\partial r} & \frac{\partial u}{\partial s} \\ \frac{\partial v}{\partial r} & \frac{\partial v}{\partial s} \end{vmatrix} \times \begin{vmatrix} \frac{\partial r}{\partial x} & \frac{\partial r}{\partial y} \\ \frac{\partial s}{\partial x} & \frac{\partial s}{\partial y} \end{vmatrix}$$

$$= \begin{vmatrix} \frac{\partial u}{\partial r} & \frac{\partial u}{\partial s} \\ \frac{\partial v}{\partial r} & \frac{\partial v}{\partial s} \end{vmatrix} \begin{vmatrix} \frac{\partial r}{\partial x} & \frac{\partial s}{\partial x} \\ \frac{\partial r}{\partial y} & \frac{\partial s}{\partial y} \end{vmatrix} = \begin{vmatrix} \frac{\partial u}{\partial r}\frac{\partial r}{\partial x} + \frac{\partial u}{\partial s}\frac{\partial s}{\partial x} & \frac{\partial u}{\partial r}\frac{\partial r}{\partial y} + \frac{\partial u}{\partial s}\frac{\partial s}{\partial y} \\ \frac{\partial v}{\partial r}\frac{\partial r}{\partial x} + \frac{\partial v}{\partial s}\frac{\partial s}{\partial x} & \frac{\partial v}{\partial r}\frac{\partial r}{\partial y} + \frac{\partial v}{\partial s}\frac{\partial s}{\partial y} \end{vmatrix}$$

$$= \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} = \frac{\partial(u, v)}{\partial(x, y)}.$$

$$\frac{\partial(u, v)}{\partial(r, s)} \times \frac{\partial(r, s)}{\partial(x, y)} = \frac{\partial(u, v)}{\partial(x, y)}.$$

Note: *If $u = u(r, s, t)$, $v = v(r, s, t)$, and $w = w(r, s, t)$ are functions of three variables r, s and t, where $r = r(x, y, z)$, $s = s(x, y, z)$ be the functions of three variables x, y and z, then*

$$\frac{\partial(u, v, w)}{\partial(x, y, z)} = \frac{\partial(u, v, w)}{\partial(r, s, t)} \times \frac{\partial(r, s, t)}{\partial(x, y, z)}$$

Property 3. *If $f(u, v) = 0$, and $u = u(x, y)$ and $v = v(x, y,)$ be the functions of two variables, then* $\dfrac{\partial(u, v)}{\partial(x, y)} = 0$.

Proof: Here, $u = u(x, y)$ and $v = v(x, y)$, and the function is given by

$$f(u, v) = 0. \qquad \ldots(1)$$

Differentiating with respect to x, and y respectively, we get

$$\frac{\partial f}{\partial u}\frac{\partial u}{\partial x} + \frac{\partial f}{\partial v}\frac{\partial v}{\partial x} = 0, \qquad \ldots(2)$$

$$\frac{\partial f}{\partial u}\frac{\partial u}{\partial y} + \frac{\partial f}{\partial v}\frac{\partial v}{\partial y} = 0. \qquad \ldots(3)$$

Eliminating $\partial f/\partial u$, and $\partial f/\partial v$ from (2), and (3), we get

$$\begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial v}{\partial x} \\ \frac{\partial u}{\partial y} & \frac{\partial v}{\partial y} \end{vmatrix} = 0.$$

On interchanging rows and columns, we get

$$\begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} = 0,$$

$$\therefore \quad \frac{\partial(u, v)}{\partial(x, y)} = 0.$$

Note: *If $f(u, v, w) = 0$, and $u = u(x, y, z)$ and $v = v(x, y, z)$, and $w = w(x, y, z)$ be the functions of three variables, then* $\frac{\partial(u, v, w)}{\partial(x, y, z)} = 0$.

In engineering and data science, the Jacobian determinant plays a vital role in analyzing transformations and system stability. In robotics, it links joint velocities to end-effector velocities, facilitating kinematic control. In finite element analysis (FEA), it maps local element coordinates to global coordinates, crucial for accurate integration. In control systems, it linearizes nonlinear systems and evaluates stability. In data science, the Jacobian is essential for optimization algorithms such as gradient descent, backpropagation in neural networks, and normalizing flows in generative models, ensuring correct data transformation and likelihood computation. Its importance is also evident in dimensionality reduction and Bayesian inference, highlighting its significance in modeling and analysis tasks.

Worked out Examples

1. Verify Euler's theorem for the function $u = f(x, y) = ax^2 + 2hxy + by^2$

Solution: A function u of two variables x, y, is given by

$$u = f(x, y) = ax^2 + 2hxy + by^2. \qquad \ldots(1)$$

Replacing x by tx, and y by ty in (1), where t is any variable, we get

$$f(tx, ty) = a(tx)^2 + 2h(tx)(ty) + b(ty)^2 = at^2x^2 + 2ht^2xy + bt^2y^2$$
$$= t^2(ax^2 + 2hxy + by^2) = t^2 f(x, y).$$

Hence, the function u is homogenous of degree two. We have to verify that the Euler's theorem for

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 2u.$$

Partially differentiating with respect to x and y to (1) respectively, we get

$$\frac{\partial u}{\partial x} = 2ax + 2hy, \frac{\partial u}{\partial y} = 2hx + 2by.$$

Now

$$x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y} = x(2ax+2hy)+y(2hx+2by) = 2ax^2+2hxy+2hxy+2b^2y^2$$
$$= 2(ax^2+2hxy+by^2) = 2u$$

$$\therefore \quad x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y} = 2\,u.$$

Hence Euler's theorem is verified.

2. Verify Euler's theorem of the function $u=\dfrac{x^{1/4}+y^{1/4}}{x^{1/5}+y^{1/5}}$

Solution: A function u of two variables x, y, is given by

$$u(x, y) = \frac{x^{1/4}+y^{1/4}}{x^{1/5}+y^{1/5}}. \qquad ...(1)$$

Replacing x by tx, and y by ty in (1), where t is any variable, we get

$$u(tx, ty) = \frac{t^{1/4}x^{1/4}+t^{1/4}y^{1/4}}{t^{1/5}x^{1/5}+t^{1/5}y^{1/5}} = \frac{t^{1/4}(x^{1/4}+y^{1/4})}{t^{1/5}(x^{1/5}+y^{1/5})}$$
$$= t^{1/20}\frac{x^{1/4}+y^{1/4}}{x^{1/5}+y^{1/5}} = t^{1/20}\,u(x, y).$$

Hence, the function is homogenous of degree 1/20. We have to verify that Euler's theorem for

$$x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y} = \frac{1}{20}u.$$

Partially differentiating (1) with respect to *x*, we get

$$\frac{\partial u}{\partial x} = \frac{(x^{1/5}+y^{1/5})(1/4)x^{-3/4}-(x^{1/4}+y^{1/4})(1/5)x^{-4/5}}{(x^{1/5}+y^{1/5})^2}.$$

Partially differentiating (1) with respect to *y*, we get

$$\frac{\partial u}{\partial y} = \frac{(x^{1/5}+y^{1/5})(1/4)y^{-3/4}-(x^{1/4}+y^{1/4})(1/5)y^{-4/5}}{(x^{1/5}+y^{1/5})^2}.$$

Now

$$x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y} = x\,\frac{(x^{1/5}+y^{1/5})(1/4)x^{-3/4}-(x^{1/4}+y^{1/4})(1/5)x^{-4/5}}{(x^{1/5}+y^{1/5})^2}$$
$$+\,y\,\frac{(x^{1/5}+y^{1/5})(1/4)y^{-3/4}-(x^{1/4}+y^{1/4})(1/5)y^{-4/5}}{(x^{1/5}+y^{1/5})^2}$$
$$= \frac{(x^{1/5}+y^{1/5})(1/4)x^{1/4}-(x^{1/4}+y^{1/4})(1/5)x^{1/5}}{(x^{1/5}+y^{1/5})^2}$$
$$+\,\frac{(x^{1/5}+y^{1/5})(1/4)y^{1/4}-(x^{1/4}+y^{1/4})(1/5)y^{1/5}}{(x^{1/5}+y^{1/5})^2}$$
$$= \frac{(1/4)(x^{1/5}+y^{1/5})(x^{1/4}+y^{1/4})-(1/5)(x^{1/4}+y^{1/4})(x^{1/5}+y^{1/5})}{(x^{1/5}+y^{1/5})^2}$$
$$= \frac{1}{20}\,\frac{(x^{1/5}+y^{1/5})(x^{1/4}+y^{1/4})}{(x^{1/5}+y^{1/5})^2} = \frac{1}{20}\,\frac{x^{1/4}+y^{1/4}}{x^{1/5}+y^{1/5}} = \frac{1}{20}u.$$

$$\therefore \quad x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y} = \frac{1}{20}u.$$

Hence Euler's theorem is verified.

3. If $u = \sin^{-1}\dfrac{x^2+y^2}{x+y}$, prove that $x\dfrac{\partial u}{\partial x} + y\dfrac{\partial u}{\partial y} = \tan u$

Solution: A function u of two variables x, y, is given by

$$u = \sin^{-1}\frac{x^2+y^2}{x+y}.$$

So
$$\sin u = \frac{x^2+y^2}{x+y} = \frac{x^2\left(1+\dfrac{y^2}{x^2}\right)}{x\left(1+\dfrac{y}{x}\right)} = x\, f\left(\frac{y}{x}\right).$$

Hence, sinu is a homogenous function of degree one, so applying Euler's theorem as

$$x\frac{\partial(\sin u)}{\partial x} + y\frac{\partial(\sin u)}{\partial y} = 1.\ \sin u,$$

or
$$x\frac{\partial(\sin u)}{\partial u}\frac{\partial u}{\partial x} + y\frac{\partial(\sin u)}{\partial u}\frac{\partial u}{\partial y} = \sin u,$$

or
$$x\cos u\frac{\partial u}{\partial x} + y\cos u\frac{\partial u}{\partial y} = \sin u,$$

$\therefore$
$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \tan u.$$

4. If $u = \log\dfrac{x^2+y^2}{x+y}$, then show that $x\dfrac{\partial u}{\partial x} + y\dfrac{\partial u}{\partial y} = 1$

Solution: A function u of two variables x and y, is given by

$$u = \log\frac{x^2+y^2}{x+y},$$

or
$$e^u = \frac{x^2+y^2}{x+y} = \frac{x^2\left(1+\dfrac{y^2}{x^2}\right)}{x\left(1+\dfrac{y}{x}\right)} = x\, f\left(\frac{y}{x}\right).$$

Hence, e^u is a homogenous function of degree one, so applying Euler's theorem as

$$x\frac{\partial(e^u)}{\partial x} + y\frac{\partial(e^u)}{\partial y} = 1.\ e^u,$$

or
$$x\frac{\partial(e^u)}{\partial u}\frac{\partial u}{\partial x} + y\frac{\partial(e^u)}{\partial u}\frac{\partial u}{\partial y} = e^u,$$

or
$$x\, e^u\frac{\partial u}{\partial x} + y\, e^u\frac{\partial u}{\partial y} = e^u,$$

$\therefore$
$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 1.$$

5. Find $\frac{du}{dt}$ if $u = \sin\left(\frac{x}{y}\right)$, $x = e^t$, $y = t^2$

Solution: A function u of two variables x and y, is given by

$$u = \sin\left(\frac{x}{y}\right), \quad \ldots(1)$$

where $\quad x = e^t, \quad y = t^2. \quad \ldots(2)$

Partially differentiating (1) with respect to x and y respectively, we get

$$\frac{\partial u}{\partial x} = \frac{1}{y}\cos\left(\frac{x}{y}\right) \text{ and } \frac{\partial u}{\partial y} = -\frac{x}{y^2}\cos\left(\frac{x}{y}\right).$$

Differentiating (2) with respect to t, we get

$$\frac{dx}{dt} = e^t, \quad \text{and} \quad \frac{dy}{dt} = 2t.$$

We have

$$\frac{du}{dt} = \frac{\partial u}{\partial x}\frac{dx}{dt} + \frac{\partial u}{\partial y}\frac{dy}{dt} = \frac{1}{y}\cos\left(\frac{x}{y}\right)e^t + \left(\frac{-x}{y^2}\right)\cos\left(\frac{x}{y}\right)2t$$

$$= \frac{e^t}{t^2}\cos\left(\frac{e^t}{t^2}\right) - \frac{2e^t}{t^3}\cos\left(\frac{e^t}{t^2}\right) = \frac{(t-2)}{t^3}e^t\cos\left(\frac{e^t}{t^2}\right)$$

$$\therefore \quad \frac{du}{dt} = \frac{(t-2)}{t^3}e^t\cos\left(\frac{e^t}{t^2}\right).$$

6. Find the total derivative $\frac{du}{dx}$ where $u = x\log(xy)$ and $x^3 + y^3 + 3xy = 1$

Solution: An implicit function f(x, y) of two variables x and y, is given by

$$f(x, y) = x^3 + y^3 + 3xy - 1 = 0.$$

Differentiating it partially with respect to x and y respectively, we get

$$f_x = 3x^2 + 3y, \text{ and } f_y = 3y^2 + 3x.$$

Total derivative of the implicit function is defined by

$$\frac{dy}{dx} = -\frac{f_x}{f_y} = -\frac{3(x^2+y)}{3(y^2+x)} = -\frac{x^2+y}{x+y^2}.$$

A function u of two variables x and y, is given by

$$u = x\log(xy).$$

Partially differentiating it with respect to x and y respectively, we get

$$\frac{\partial u}{\partial x} = 1 + \log(xy), \text{ and } \quad \frac{\partial u}{\partial y} = \frac{x}{y}.$$

Total derivative of u is defined as

$$\frac{du}{dx} = \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y}\frac{dy}{dx} = 1 + \log(xy) + \frac{x}{y}\cdot\left(-\frac{x^2+y}{x+y^2}\right) = 1 + \log xy - \frac{x(x^2+y)}{y(x+y^2)}.$$

7. Find $\frac{dy}{dx}$ by using partial derivatives if $x^y = y^x$

Solution: An implicit function f(x, y) of two variables x and y, is given by

$$f(x, y) = x^{y} - y^{x} = 0.$$

Partially differentiating it with respect to *x* and *y* respectively, we get

$$f_x = y\,x^{y-1} - y^{x}\log y, \text{ and } \quad f_y = x^y \log x - x\,y^{x-1}.$$

Total derivative of y is defined by

$$\frac{dy}{dx} = -\frac{f_x}{f_y} = -\frac{yx^{y-1} - y^x \log y}{x^y \log x - xy^{x-1}} = \frac{y}{x}\,\frac{(y - x\log x)}{(x - y\log y)}$$

8. If $z = f(x, y)$ and $x = e^u + e^{-v}$, $y = e^{-u} - e^v$, then prove that $\frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} = x\frac{\partial z}{\partial x} - y\frac{\partial z}{\partial y}$

Solution: The functions x = x(u, v) and y = y (u, v) of two variables u and v, are given by

$$x = e^u + e^{-v}, \text{ and } y = e^{-u} - e^{v}. \qquad \ldots(1)$$

Partially differentiating (1) with respect to *u* considering *v* as constant, we get

$$\frac{\partial x}{\partial u} = e^{u}, \text{ and } \qquad \frac{\partial y}{\partial u} = -e^{-u}.$$

Partially differentiating (1) with respect to *v* considering *u* as constant, we get

$$\frac{\partial x}{\partial v} = -e^{-v}, \text{ and } \qquad \frac{\partial y}{\partial v} = -e^{v}.$$

A function z of two variables x and y, is given by

$$z = f(x, y). \qquad \ldots(2)$$

Partially differentiating (2) with respect to *u*, we get

$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial u} = e^u\frac{\partial z}{\partial x} - e^{-u}\frac{\partial z}{\partial y}. \qquad \ldots(3)$$

Partially differentiating (2) with respect to *v*, we get

$$\frac{\partial z}{\partial v} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial v} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial v} = -e^{-v}\frac{\partial z}{\partial x} - e^{v}\frac{\partial z}{\partial y}. \qquad \ldots(4)$$

Subtracting (4) from (3), we get

$$\frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} = e^u\frac{\partial z}{\partial x} - e^{-u}\frac{\partial z}{\partial y} - \left(-e^{-v}\frac{\partial z}{\partial x} - e^{v}\frac{\partial z}{\partial y}\right) = e^u\frac{\partial z}{\partial x} - e^{-u}\frac{\partial z}{\partial y} + e^{-v}\frac{\partial z}{\partial x} + e^{v}\frac{\partial z}{\partial y}$$

$$= (e^u + e^{-v})\frac{\partial z}{\partial x} - (e^{-u} - e^{v})\frac{\partial z}{\partial y} = u\frac{\partial z}{\partial x} - v\frac{\partial z}{\partial y}$$

$$\therefore \quad \frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} = u\frac{\partial z}{\partial x} - v\frac{\partial z}{\partial y}.$$

9. If $u = x^2$ and $v = y^2$, then find $\frac{\partial(u, v)}{\partial(x, y)}$

Solution: Here, $u = x^2$ and $v = y^2$, then the partial derivatives are

$$\frac{\partial u}{\partial x} = 2x, \frac{\partial u}{\partial y} = 0, \frac{\partial v}{\partial x} = 0, \text{ and } \frac{\partial v}{\partial y} = 2y.$$

The Jacobian of u and v with respect to x and y is given by

$$\frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} = \begin{vmatrix} 2x & 0 \\ 0 & 2y \end{vmatrix} = 4xy.$$

10. If $x = r\cos\theta$ and $y = r\sin\theta$, then prove that $\frac{\partial(x, y)}{\partial(r, \theta)} = r$ and $\frac{\partial(r, \theta)}{\partial(x, y)} = \frac{1}{r}$

Solution: Here, $x = r\cos\theta$ and $y = r\sin\theta$, then the partial derivatives are

$$\frac{\partial x}{\partial r} = \cos\theta, \frac{\partial x}{\partial \theta} = -r\sin\theta, \frac{\partial y}{\partial r} = \sin\theta, \text{ and } \frac{\partial y}{\partial \theta} = r\cos\theta.$$

The Jacobian of x and y with respect to r and θ is given by

$$\frac{\partial(x, y)}{\partial(r, \theta)} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix} = r(\cos^2\theta + \sin^2\theta) = r.$$

The function can be written as $r^2 = x^2 + y^2$, and $\theta = \tan^{-1}(y/x)$, then the partial derivatives are

$$\frac{\partial r}{\partial x} = \frac{x}{r} = \frac{r\cos\theta}{r} = \cos\theta, \quad \frac{\partial r}{\partial y} = \frac{y}{r} = \frac{r\sin\theta}{r} = \sin\theta,$$

$$\frac{\partial \theta}{\partial x} = -\frac{y}{x^2+y^2} = -\frac{r\sin\theta}{r^2} = -\frac{\sin\theta}{r}, \frac{\partial \theta}{\partial y} = \frac{x}{x^2+y^2} = \frac{r\cos\theta}{r^2} = \frac{\cos\theta}{r},$$

and the Jacobian of r and θ with respect to u and v is given by

$$\frac{\partial(r, \theta)}{\partial(x, y)} = \begin{vmatrix} \frac{\partial r}{\partial x} & \frac{\partial r}{\partial y} \\ \frac{\partial \theta}{\partial x} & \frac{\partial \theta}{\partial y} \end{vmatrix} = \begin{vmatrix} \cos\theta & \sin\theta \\ -\frac{\sin\theta}{r} & \frac{\cos\theta}{r} \end{vmatrix} = \frac{1}{r}(\cos^2\theta + \sin^2\theta) = \frac{1}{r}.$$

11. If $u = xyz$ and $v = xy + yz + zx$, $w = x + y + z$, then find $\frac{\partial(u, v, w)}{\partial(x, y, z)}$

Solution: Here, $u = xyz$, $v = xy + yz + zx$, and $w = x + y + z$, then the partial derivatives are

$$\frac{\partial u}{\partial x} = yz, \frac{\partial u}{\partial y} = zx, \frac{\partial u}{\partial z} = xy, \frac{\partial v}{\partial x} = y + z, \frac{\partial v}{\partial y} = z + x, \frac{\partial v}{\partial z} = x + y,$$

$$\frac{\partial w}{\partial x} = 1, \frac{\partial w}{\partial y} = 1, \text{ and } \frac{\partial v}{\partial z} = 1.$$

The Jacobian of u, v and w with respect to x, y and z is given by

$$\frac{\partial(u, v, w)}{\partial(x, y, z)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{vmatrix} = \begin{vmatrix} yz & zx & xy \\ y+z & z+x & x+y \\ 1 & 1 & 1 \end{vmatrix}.$$

Applying $C_2 \to C_2 - C_1$, $C_3 \to C_3 - C_1$, we get

$$= \begin{vmatrix} yz & z(x-y) & y(x-z) \\ y+z & x-y & x-z \\ 1 & 0 & 0 \end{vmatrix} = \begin{vmatrix} z(x-y) & y(x-z) \\ x-y & x-z \end{vmatrix}.$$

Taking common (x-y) in first column, and (x -z) in the second column, we get

$$\frac{\partial(u, v, w)}{\partial(x, y, z)} = -(x-y)(z-x)\begin{vmatrix} z & y \\ 1 & 1 \end{vmatrix} = (x-y)(y-z)(z-x).$$

12. If $x = r\cos\theta\sin\phi$, $y = r\sin\theta\sin\phi$, $z = r\cos\phi$, then prove that $\frac{\partial(x, y, z)}{\partial(r, \theta, \phi)} = r^2\sin\phi$

Solution: Here, $x = r\cos\theta\sin\phi$, $y = r\sin\theta\sin\phi$, $z = r\cos\phi$, then the partial derivatives are

$$\frac{\partial x}{\partial r} = \cos\theta\sin\phi,\quad \frac{\partial x}{\partial \theta} = -r\sin\theta\sin\phi,\quad \frac{\partial x}{\partial \phi} = r\cos\theta\cos\phi,\quad \frac{\partial y}{\partial r} = \sin\theta\sin\phi,$$

$$\frac{\partial y}{\partial \theta} = r\cos\theta\sin\phi,\quad \frac{\partial y}{\partial \phi} = r\sin\theta\cos\phi,\quad \frac{\partial z}{\partial r} = \cos\phi,\quad \frac{\partial z}{\partial \theta} = 0, \text{ and } \frac{\partial z}{\partial \phi} = -r\sin\phi.$$

The Jacobian of x, y and z with respect to r, ϕ and θ is given by

$$\frac{\partial(x, y, z)}{\partial(r, \phi, \theta)} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \phi} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \phi} & \frac{\partial y}{\partial \theta} \\ \frac{\partial z}{\partial r} & \frac{\partial z}{\partial \phi} & \frac{\partial z}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \cos\theta\sin\phi & r\cos\theta\cos\phi & -r\sin\theta\sin\phi \\ \sin\theta\sin\phi & r\sin\theta\cos\phi & r\cos\theta\sin\phi \\ \cos\phi & -r\sin\phi & 0 \end{vmatrix}.$$

Expanding in R_3, we get

$$= \cos\phi\,(r^2\sin\phi\cos\phi)(\cos^2\theta + \sin^2\theta) + r\sin\phi\sin^2\phi\,(\cos^2\theta + \sin^2\theta)$$

$$= r^2\sin\phi\cos^2\phi + r\sin\phi\sin^2\phi = r^2\sin\phi\,(\cos^2\phi + \sin^2\phi) = r^2\sin^2\phi.$$

$$\therefore\ \frac{\partial(x, y, z)}{\partial(r, \phi, \theta)} = r^2\sin^2\phi.$$

Formula to find Error/ Change.

Exercise - 2

1. Verify Euler's theorem for the following functions

(i) $u = x f\left(\frac{y}{x}\right)$ (ii) $u = \sin^{-1}\frac{x}{y} + \tan^{-1}\frac{y}{x}$

(iii) $u = x^n \tan^{-1}\left(\frac{y}{x}\right)$ (iv) $u = x^n \sin\left(\frac{y}{x}\right)$

(v) $u = \frac{x^2 y^2}{x^3 + y^3}$ (vi) $u = \frac{x^3 y}{x^2 + y^2}$

2. (i) If $u = \sin^{-1}\frac{x^2y^2}{x+y}$, then show that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 3\tan u$

(ii) If $u = \log(x^2y)$, then show that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 3$

(iii) If $\sin u = \frac{\sqrt{x} - \sqrt{y}}{\sqrt{x} + \sqrt{y}}$, then show that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 0$

(iv) If $u = \tan^{-1}\frac{x^3 + y^3}{x - y}$, then show that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \sin 2u$

(v) If $u = \operatorname{cosec}^{-1}\left(\frac{x^{1/2} + y^{1/2}}{x^{1/3} + y^{1/3}}\right)$, then show that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = -\frac{1}{6}\tan u$

(vi) If $u = \cos^{-1}\left(\frac{x+y}{\sqrt{x}+\sqrt{y}}\right)$, then show that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = -\frac{1}{2}\cot u$

(vii) If $u = \log(x^2 + y^2 + z^2)$, then prove that $x\frac{\partial^2 u}{\partial y\,\partial z} = y\frac{\partial^2 u}{\partial z\,\partial x}$

3. Find $\frac{dy}{dx}$ in the following functions

(i) $x^{2/3} + y^{2/3} = 4$ (ii) $2x^3 + x^2y + y^3 = 1$

(iii) $(\tan x)^y + (y)^{\tan x} = 0$ (iv) $x^3 + y^3 = 3axy$

4. Find $\frac{dz}{dt}$ in the following functions

(i) $z = x^3 - y^3$ where $x = \frac{1}{t+1}$, $y = \frac{t}{t+1}$

(ii) $z = \log(x + y)$ where $x = e^{-2t}$, $y = t^3 - t^2$

(iii) $z = \sin^{-1}(x - y)$ where $x = 3t$, $y = 4t^3$

Error and Approximations.
Error (Change)

5. Find $\frac{dz}{dx}$ in the following functions

(i) $z = x\cos y + y\sin x$ where $y = x^2 + 1$ (ii) $z = (y + x)e^{xy}$ where $y = \frac{1}{x^2}$

6. If $u = f(x, y) = x^2 - 2xy + 3y$, find the differential du and the change Δu in u if f(x, y) changes

from (1, 2) to (1.03, 1.99)

7. If $u = f(x, y) = x^3 + y^3$ find the differential du and change Δu in u if f(x, y) changes from (10, 10) to (10.1, 10.1).

8. If $u = f(x, y) = x^2 - 3x^3y^2 + 4x - 2y^3 + 6$. Find the differential du and change Δu in u if f(x, y) changes from (-2, 3) to (-2. 02, 3.01)

9. The radius and altitude of a rigid circular cylinder are measured as 3 inches and 8 inches respectively with possible error in measurement of 0.05 inches. Find the maximum error in the calculated volume of the cylinder

10. If the dimensions in inches of a rectangular box change from 9, 6, and 4 to 9.02, 5.97 and 4.01 respectively. Find the change in volume

11. If $u = x + y$ and $v = xy^2$, then show that $\dfrac{\partial(u, v)}{\partial(x, y)} = 2xy - y^2$.

12. If $x = u - v$ and $y = 2uv$, then show that $\dfrac{\partial(x, y)}{\partial(u, v)} = 2(u + v)$

13. If $u = \dfrac{yz}{x}$ and $v = \dfrac{zx}{y}$, and $w = \dfrac{xy}{z}$, then show that $\dfrac{\partial(u, v, w)}{\partial(x, y, z)} = 4$.

14. If $x = u + v + w$ and $y = vw + wu + uv$, and $z = uvw$, then show that

$$\frac{\partial(x, y, z)}{\partial(u, v, w)} = -(u - v)(v - w)(w - u)$$

$dx = \Delta x$
$dy = \Delta y$
$du = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy$

Answers

$\Delta u = f(x + \Delta x, y + \Delta y) - f(x, y)$

3. (i) $-\left(\dfrac{y}{x}\right)^{1/3}$. (ii) $-\left(\dfrac{2xy + 6x^2}{x^2 + 3y^2}\right)$ (iii) $-\dfrac{\sec^2 x\,[y(\tan x)^{y-1} + \log y.\, y^{\tan x}]}{(\tan x)^y \log(\tan x) + \tan x\, y^{\tan x - 1}}$ (iv) $\dfrac{x^2 - ay}{ax - y^2}$.

4. (i) $\dfrac{-3(t^2 + 1)}{(t + 1)^4}$ (ii) $\dfrac{3t^2 - 2t - 2e^{-2t}}{t^3 - t^2 + e^{-2t}}$ (iii) $\dfrac{3}{(1 - t^2)^{1/2}}$

5. (i) $\cos(x^2 + 1) + (x^2 + 1)\cos x - 2x^2 \sin(x^2 + 1) + 2x \sin x$. (ii) $e^{1/x}\left[1 - \dfrac{1}{x} - \dfrac{2}{x^3} - \dfrac{1}{x^4}\right]$

6. - 0.070, - 0.039 7. 600, 60.602 8. 7.38, 7.490 9. 95 inch^3 10. - 0.063906

1.2 Extreme Values of Two and More Variables

We have previously discussed the theory of extreme values for functions of one variable. Now, we delve into the theory of functions of two or three variables. A maximum or minimum value of a function of two or three independent variables is known as an extreme value of the function. It concerns the evaluation of the maximum and minimum values of the function.

A function f(x, y) is said to have a *maximum* or *minimum* at $x = a$, $y = b$ according to

$$f(a + h, b + k) < f(a, b), \quad \text{or } f(a + h, b + k) > f(a, b)$$

for all positive or negative small values of h and k.

1.2.1 Conditions for f(x, y) to be Maximum or Minimum

Theorem 4 *The necessary conditions for $f(x, y)$ to have an extreme value at (a, b) is that $f_x(a, b) = 0$, $f_y(a, b) = 0$ provided these partial derivatives exist*

Proof: If f(a, b) is an extreme value of the function f(x, y) of two variables, then it must also be an extreme value of both functions f(x, b) and f(a, y) of one variable. The necessary conditions for that f(x, b) and f(a, y) have extreme values at $x = a$ and $y = b$ respectively, which is

$$f_x(a, b) = 0, \quad f_y(a, b) = 0.$$

If these conditions are satisfied, then for small value of h and k, by applying Taylor 's theorem, we get

$$(a + h, b + k) - f(a, b) = \left(h\frac{\partial f}{\partial x} + k\frac{\partial f}{\partial y}\right)\text{at}(a, b) + \frac{1}{2!}\left(h^2\frac{\partial^2 f}{\partial x^2} + 2hk\frac{\partial^2 f}{\partial x\partial y} + k^2\frac{\partial^2 f}{\partial y^2}\right) + \ldots$$

$$= \frac{1}{2!}(h^2r + 2hks + k^2t) \text{ where } r = f_{xx}(a, b), s = f_{xy}(a, b) \text{ and } t = f_{yy}(a, b)$$

$$= \frac{1}{2r}[(hr + ks)^2 + k^2(rt - s^2)].$$

Since $(hr + ks)^2$ is always positive and $k^2(rt - s^2)$ will be positive if $rt - s^2 > 0$, it follows that if $rt - s^2 > 0$, then f(x, y) has a *maximum* or a *minimum* at (a, b) according to whether $r < 0$ or $r > 0$. If $rt - s^2 < 0$, then f(x, y) has no maximum or minimum at (a, b); in this case, the point is called a *saddle point*. If $rt - s^2 = 0$, then the further investigation is necessary.

Stationary value: The value f(a, b) is called the *stationary value* of the function f(x, y) if

$$f_x(a, b) = 0 \quad \text{and} \quad f_y(a, b) = 0,$$

making the function stationary at (a, b). Every extreme value is a stationary value but the converse may not be true. Stationary points that are not extreme points are called the *saddle points*.

Procedure for determining maximum and minimum values of f(x, y)

(i) Find $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial^2 f}{\partial x^2}, \frac{\partial^2 f}{\partial x\partial y}, \frac{\partial^2 f}{\partial y^2},$

(ii) Solve $\frac{\partial f}{\partial x} = 0$ and $\frac{\partial f}{\partial y} = 0$ simultaneously in x and y to get $(x, y) = (a, b), (c, d)$.....

(iii) For each solution, find $r = \frac{\partial^2 f}{\partial x^2}$, $s = \frac{\partial^2 f}{\partial x\partial y}$, $t = \frac{\partial^2 f}{\partial y^2}$,

(iv) If $rt - s^2 > 0$ and $r < 0$, then f(x, y) has a maximum value. If $rt - s^2 < 0$, then f(x, y) has no extreme value. If $rt - s^2 = 0$, further investigation is necessary.

The function f(x, y) of two variables x and y is said to have an extreme value f(a, b) if

$$f_x(a, b) = 0, f_y(a, b) = 0 \text{ and } f_{xx}(a, b) - f_{yy}(a, b) - [f_{xy}(a, b)]^2 > 0.$$

And this extreme value is maximum or minimum according to

$$f_{xx}(a, b) < 0 \text{ or } \quad f_{xx}(a, b) > 0.$$

Further investigation is necessary if

$$f_{xx}(a, b)\, f_{yy}(a, b) - [f_{xy}(a, b)]^2 = 0.$$

Definition 1.3 *A function f(x, y, z) is said to have a maximum or minimum at x = a, y = b, z = c according as* $f(a + h, b + k, c + l) < f(a, b, c)$

or $f(a + h, b + k, c + l) > f(a, b, c)$ *for all positive or negative small values of h, k and l.*

In other words, to difference $f(a + h, b + k, c + l) - f(a, b, c)$ is of the same sign for all small values of h, k and l.

If the sign of difference is negative then f(a, b, c) is a maximum and if this sign of difference is positive f(a, b, c) is a minimum.

1.2.2 Necessary and Sufficient Conditions

Theorem 5 *The necessary condition for f(a, b, c) to be minimum or maximum value of the function f are that all the partial derivatives* f_x, f_y, f_z *exist and vanish at (a, b, c)*

Proof: A point (a, b, c) is called a *Stationary point* if all the first order partial derivatives of the function vanish at that point. Thus, the stationary points are determined by solving the following three equations simultaneously.

$$f_x(x, y, z) = 0, \qquad f_y(x, y, z) = 0, \qquad f_z(x, y, z) = 0.$$

The necessary sufficient conditions for a function f(x, y, z) of three independent variables to have *maximum* value at (a, b, c) are that

$$f_x = 0, \qquad f_y = 0, \qquad f_z = 0,$$

$$f_{xx} < 0, \quad \begin{vmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{vmatrix} > 0, \quad \begin{vmatrix} f_{xx} & f_{xy} & f_{xz} \\ f_{yx} & f_{yy} & f_{yz} \\ f_{zx} & f_{zy} & f_{zz} \end{vmatrix} < 0,$$

and *minimum* value at (a, b, c) are that

$$f_x = 0, \qquad f_y = 0, \qquad f_z = 0,$$

$$f_{xx} > 0, \quad \begin{vmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{vmatrix} > 0, \quad \begin{vmatrix} f_{xx} & f_{xy} & f_{xz} \\ f_{yx} & f_{yy} & f_{yz} \\ f_{zx} & f_{zy} & f_{zz} \end{vmatrix} > 0.$$

Worked out Examples

1. Examine the maxima and minima of the function $f(x, y) = x^3 + y^3 - 3x - 12y + 20$

Solution: A function f(x,y) of two variables x and y, is given by

$$f(x, y) = x^3 + y^3 - 3x - 12y + 20.$$

Differentiating it partially with respect to x, and y respectively, we get

$$f_x = 3x^2 - 3, \quad f_y = 3y^2 - 12.$$

For the stationary points, putting

$$f_x = 0, \quad \text{and} \quad f_y = 0,$$

$$\therefore \quad x = \pm 1 \text{ and } y = \pm 2.$$

Thus, the function has four stationary points: (1, 2), (- 1, 2), (1, - 2), (- 1, - 2).

Now $\quad f_{xx} = 6x, \quad f_{xy} = 0, \quad f_{yy} = 6y$

At (1, 2) $\quad f_{xx} = 6 >$ and $f_{xx}\, f_{yy} - (f_{xy})^2 = 72 > 0.$

Hence (1, 2) is a point of minima of the function .

At (- 1, 2), $\quad f_{xx} = -6 < 0$ and $f_{xx}\, f_{yy} - (f_{xy})^2 = -72 < 0.$

Hence the function has neither maximum nor minimum at (- 1, 2).

At (1, - 2), $\quad f_{xx} = 6 > 0$ and $f_{xx}\, f_{yy} - (f_{xy})^2 = -72 < 0.$

Hence, the function has neither maximum nor minimum at (1, - 2).

At (- 1, -2), $\quad f_{xx} = -6 < 0$ and $f_{xx}f_{yy} - (f_{xy})^2 = 72 > 0.$

Hence the function has a maximum value at (- 1, -2). The points (- 1, 2), (1, - 2) are saddle points.

$$\text{Min. value} = f(1, 2) = 1 + 8 - 3 - 24 + 20 = 2,$$

$$\text{Max. value} = f(-1, -2) = -1 - 8 + 3 + 24 + 20 = 38.$$

2. Find the maximum and minimum values of $x^3 - x^2 - y^2 + xy$

Solution: A function f of two variables x and y, is given by

$$f(x, y) = x^3 - x^2 - y^2 + xy. \qquad \ldots(1)$$

Partially differentiating (1) with respect to x and y respectively, we get

$$f_x = 3x^2 - 2x + y, \qquad f_y = -2y + x,$$

$$f_{xx} = 6x - 2, \qquad f_{yx} = 1 = f_{xy}, \quad f_{yy} = -2.$$

For the extreme values, putting $f_x = 0$, and $f_y = 0$, we get

$$3x^2 - 2x + y = 0, \text{ and } -2y + x = 0.$$

Solving these two equations, we get

$$x = 0,\ \frac{1}{2}; \quad y = 0,\ \frac{1}{4}.$$

Therefore the stationary points are (0, 0) and $\left(\frac{1}{2}, \frac{1}{4}\right)$.

At (0, 0), $\qquad f_{xx} = -2 < 0, \quad f_{xx}\, f_{yy} - f_{xy}^{\ 2} = (-2) \times (-2) - 1 = 3 > 0.$

Hence, f(x, y) has maximum value at (0, 0). Max. value = f(0, 0) = 0 - 0 + 0 + 0 = 0.

At $\left(\frac{1}{2}, \frac{1}{4}\right)$, $\qquad f_{xx} = 1 > 0, \quad f_{xx}\, f_{yy} - f_{xy}^{\ 2} = -3 < 0.$

Hence, f has neither maximum nor minimum. So, the points $\left(\frac{1}{2}, \frac{1}{4}\right)$ is saddle point.

3. Find the maximum and minimum values of $xy + \frac{a^3}{x} + \frac{a^3}{y}$

Solution: A function f of two variables x and y, is given by

$$f(x, y) = xy + \frac{a^3}{x} + \frac{a^3}{y}. \qquad \ldots(1)$$

Partially differentiating (1) with respect to x and and y respectively, we get

$$f_x = y - \frac{a^3}{x^2}, \qquad f_y = x - \frac{a^3}{y^2}, \qquad f_{xx} = \frac{2a^3}{x^3},\ f_{xy} = 1, \quad f_{yy} = \frac{2a^3}{y^3}.$$

For the extreme values, putting $f_x = 0$, and $f_y = 0$, we get

$$y - \frac{a^3}{x^2} = 0, \quad \text{and} \quad x - \frac{a^3}{y^2} = 0.$$

Solving these equations $x^2y - a^3 = 0$, and $xy^2 - a^3 = 0$, we get

$$x = 0,\ y = 0 \text{ and } x = a,\ y = a.$$

Therefore, the stationary points are (0, 0) and (a, a).

At (a, a),

$$f_{xx} = 2 > 0, \qquad f_{xx}\, f_{yy} - f_{xy}^{\ 2} = 3 > 0.$$

Hence, f(x, y) has minimum value at (a, a).

$$\text{Min. value} = f(a, a) = a^2 + a^2 + a^2 = 3a^2.$$

4. Find the maximum and minimum values $x^3y^2(1 - x - y),\ x \neq 0,\ y \neq 0,\ x + y \neq 1$

Solution: A function f of two variables x and y, is given by

$$f(x, y) = x^3y^2(1 - x - y), \quad x \neq 0, y \neq 0, x + y \neq 1. \qquad \ldots(1)$$

Partially differentiating (1) with respect to x and and y respectively, we get

$$f_x = 3x^2y^2 - 4x^3y^2 - 3x^2y^3, \qquad f_y = 2x^3y - 2x^4y - 3x^3y^2,$$

$$f_{yx} = 6x^2y - 8x^3y - 9x^2y^2 = f_{xy}, \quad f_{yy} = 2x^3 - 2x^4 - 6x^3y,$$

$$f_{xx} = 6x^3y^2 - 12x^2y^2 - 6xy^3.$$

For the extreme values, puuting $f_x = 0$, and $f_y = 0$, we get

$$3x^2y^2 - 4x^3y^2 - 3x^2y^3 = 0 \text{ and } \quad 2x^3y - 2x^4y - 3x^3y^2 = 0.$$

Since $x \neq 0, y \neq 0$,

$$3 - 4x - 3y = 0 \text{ and } 2 - 2x - 3y = 0.$$

Solving these $x = \frac{1}{2}, y = \frac{1}{3}$. Therefore the stationary point is $\left(\frac{1}{2}, \frac{1}{3}\right)$.

At $\left(\frac{1}{2}, \frac{1}{3}\right)$, $f_x = \frac{-17}{36} < 0, \quad f_{xx}\, f_{yy} - f_{xy}^{\ 2} = \frac{15}{288} > 0.$

Hence, the function f(x, y) has maximum value at $\left(\frac{1}{2}, \frac{1}{3}\right)$.

$$\text{Max. value} = f\left(\frac{1}{2}, \frac{1}{3}\right) = \frac{1}{72}\left(1 - \frac{1}{2} - \frac{1}{3}\right) = \frac{1}{432}.$$

5. Find the maximum and minimum values $(x + y + z)^3 - 3(x + y + z) - 24\,xyz + 1$

Solution: A function f of three variables x, y and z, is given by

$$f(x, y, z) = (x + y + z)^3 - 3\,(x + y + z) - 24xyz + 1. \qquad \ldots(1)$$

Partially differentiating (1) with respect to *x*, *y* and z respectively, we get

$$f_x = 3(x + y + z)^2 - 3 - 24yz, \qquad f_y = 3(x + y + z)^2 - 3 - 24xz,$$

$$f_z = 3(x + y + z)^2 - 3 - 24xy, \qquad f_{xx} = 6(x + y + z),$$

$$f_{yx} = 6(x + y + z) - 24z, \qquad f_{yy} = 6(x + y + z),$$

$$f_{xz} = 6(x + y + z) - 24y, \qquad f_{zz} = 6(x + y + z), \qquad f_{yz} = 6(x + y + z) - 24x.$$

For the extreme values, putting $f_x = 0, \quad f_y = 0, \quad f_z = 0$, we get

$$(x + y + z)^2 - 1 - 8yz = 0, \quad (x + y + z)^2 - 1 - 8zx = 0,$$

and $\qquad (x + y + z)^2 - 1 - 8xy = 0.$

Solving these equations, we get

$$x = y = z, \text{ and } \quad x = 1, -1; \quad y = 1, -1; \quad z = 1, -1.$$

So the stationary points are (1, 1, 1) and (- 1, - 1, - 1).

At (1, 1, 1), $\qquad f_{xx} = 18 > 0,\ f_{xy} = 18 - 24 = -6 = f_{yx},\ f_{yy} = 18 > 0$, so

$$\begin{vmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{vmatrix} = \begin{vmatrix} 18 & -6 \\ -6 & 18 \end{vmatrix} = 324 - 36 = 288 > 0,$$

and $\qquad f_{zx} = 18 - 24 = -6,\ f_{zy} = 18 - 24 = -6 = f_{yz},\ f_{yy} = 18,\ f_{zz} = 18$, so

$$\begin{vmatrix} f_{xx} & f_{xy} & f_{xz} \\ f_{yx} & f_{yy} & f_{yz} \\ f_{zx} & f_{zy} & f_{zz} \end{vmatrix} = \begin{vmatrix} 18 & -6 & -6 \\ -6 & 18 & -6 \\ -6 & -6 & 18 \end{vmatrix} = 3456 > 0.$$

Hence, the function f(x, y) has minimum value at (1, 1, 1).

$$\text{Min. value} = (1 + 1 + 1)^3 - 3(1 + 1 + 1) - 24 + 1 = 27 - 9 - 24 + 1 = -5.$$

At (- 1, - 1, - 1), $\qquad f_{xx} = -18 < 0,\ f_{xy} = -18 + 24 = 6 = f_{yx},\ f_{yy} = -18 < 0$, so

$$\begin{vmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{vmatrix} = \begin{vmatrix} 18 & -6 \\ -6 & 18 \end{vmatrix} = 324 - 36 = 288 > 0,$$

and $\qquad f_{zx} = -18 + 24 = 6,\ f_{zy} = -18 + 24 = 6 = f_{yz},\ f_{yy} = -18,\ f_{zz} = -18$, so

$$\begin{vmatrix} f_{xx} & f_{xy} & f_{xz} \\ f_{yx} & f_{yy} & f_{yz} \\ f_{zx} & f_{zy} & f_{zz} \end{vmatrix} = \begin{vmatrix} -18 & 6 & 6 \\ 6 & -18 & 6 \\ 6 & 6 & -18 \end{vmatrix} = -3456 < 0.$$

Hence, the function f(x, y) has maximum value at (-1,- 1,- 1),

$$\text{Max. value} = (-1-1-1)^3 - 3(-1-1-1) + 24 + 1 = -27 + 9 + 24 + 1 = 7.$$

6. Show that $f(x, y) = y^2 + x^2y + x^4$ has a minimum at (0, 0)

Solution: A function f of two variables x and y, is given by

- $f(x, y) = y^2 + x^2y + x^4.$

Partially differentiating with respect to *x* and *y* respectively, we get

$$f_x = 2xy + 4x^3, \quad f_y = 2y + x^2,$$

$$f_{xx} = 2y + 12x^2, \quad f_{yy} = 2, \quad f_{xy} = 2x.$$

It can be easily verified that at (0, 0), $f_x = 0$, $f_y = 0$, $f_{xx} = 0$, $f_{yy} = 2$, $f_{xy} = 0$.

Thus at the origin, $f_{xx} f_{yy} - (f_{xy})^2 = 0$.

It is doubtful case and requires further investigation. The given function can be written as

$$f(x, y) = \left(y + \frac{x^2}{2}\right)^2 + \frac{3x^4}{4}, \text{ and } f(x, y) - f(0, 0) = \left(y + \frac{x^2}{2}\right)^2 + \frac{3x^4}{4}$$

This is greater than zero for all values of x and y. Hence, f(x, y) has minimum at (0, 0).

Stationary values under the given conditions

Sometimes it is required to find the stationary values of a function of two and three independent variables connected by some given conditions. With the help of given conditions, the function can reduce to a function of a single variable *x* or two variables *x* and *y*.

1. *Stationary points of the function $f(x, y)$ under the condition $\phi(x, y) = 0$*

A function F of two variables x and y is given by

$$F = f(x, y), \qquad ...(1)$$

under the condition

$$\phi(x, y) = 0. \qquad ...(2)$$

The function F of (1) is reduced to a function of a single variable x with the help of (2). To obtain stationary points, putting

$$\frac{dF(x)}{dx} = 0,$$

which gives the value of *x*. After substituting the value of *x* in (2), we get y and thus will be obtain the stationary points.

2. *Stationary points of the function $f(x, y, z)$ under the condition $\phi(x, y, z) = 0$*

A function F of three variables x, y and z, is given by

$$F = f(x, y, z), \quad .....(1)$$

under the condition

$$\phi(x, y, z) = 0 \quad .....(2)$$

The function F of (1) is reduced to a function of two variables x and y with the help of (2). To obtain stationary point, putting

$$F_x(x, y) = 0, F_y(x, y) = 0.$$

Solving these, we will get the value of x, and y. After substituting the values of x and y in (2), we will get z and thus will be obtained the stationary points.

3. *Stationary points of the function $f(x, y, z)$ under the conditions $\phi_1(x, y, z) = 0$, and $\phi_2(x, y, z) = 0$*

A function F of three variables x, y and z, is given by

$$F = f(x, y, z), \quad ....(1)$$

under the conditions

$$\phi_1(x, y, z) = 0, \quad ...(2)$$

and

$$\phi_2(x, y, z) = 0. \quad ....(3)$$

The function F of (1) is reduced intto a single variable x with the help of (2) and (3). To obtain stationary point, putting $\frac{dF(x)}{dx} = 0$,

and solving this, we will get the value of x. After substituting the value of x to (ii) and (iii), we will get the value of x and y, and thus will be obtained the stationary point.

Worked out Examples

1. Obtain the maximum value of xyz such that x + y + z = 24

Solution: A function f of three variables x, y and z, is given by

$$f(x, y, z) = xyz, \quad ...(1)$$

with the condition $x + y + z = 24$,

$$\text{or} \quad z = 24 - x - y. \quad ...(2)$$

With the help of (2), the function (1) can be expressed as

$$f = xy(24 - x - y) = 24xy - x^2y - xy^2. \quad ...(3)$$

Partially differentiating (3) with respect to x and y successively, we get

$$f_x = 24y - 2xy - y^2, \qquad f_y = 24x - x^2 - 2xy,$$

$$f_{xx} = -2y, \qquad f_{xy} = 24 - 2x - 2y, \qquad f_{yy} = -2x.$$

For the extreme values, putting $f_x = 0$ and $f_y = 0$, we get

$$24y - 2xy - y^2 = 0 \text{ and } 24x - x^2 - 2xy = 0,$$

$$\text{or} \quad 24 - 2x - y = 0 \text{ and } 24 - x - 2y = 0.$$

Solving these equations, we get $x = 8, y = 8$, then putting the values of x and y in (2), we get

$$z = 24 - 8 - 8 = 8.$$

Therefore, the stationary point is (8, 8, 8). At (8, 8, 8),

$$f_{xx} = -16 < 0 \text{ and } f_{xx} f_{yy} - (f_{xy})^2 = 256 - 64 = 192 > 0.$$

Hence f has maximum value at (8, 8, 8). Putting the values of x, y and z in (1), we get

$$\text{Max. value} = 8 \times 8 \times 8 = 512.$$

2. Find the minimum value of $x^2 + y^2 + z^2$ connected by the relation $ax + by + cz = p$

Solution: A function f of three variables x, y and z, is given by

$$f(x, y, z) = x^2 + y^2 + z^2, \qquad ...(1)$$

connected by the relation

$$ax + by + cz = p,$$

or $$z = \frac{(p - ax - by)}{c}. \qquad ...(2)$$

With the help of (2), the function (1) reduces to

$$f = x^2 + y^2 + \frac{1}{c^2}(p - ax - by)^2. \qquad ...(3)$$

Partially differentiating (3) with respect to x and y succesively, we get

$$f_x = 2x - \frac{2a}{c^2}(p - ax - by), \quad f_{xx} = 2 + \frac{2a^2}{c^2},$$

$$f_y = 2y - \frac{2b}{c^2}(p - ax - by), \quad f_{xy} = \frac{2ab}{c^2}, \quad f_{yy} = 2 + \frac{2b^2}{c^2}.$$

For the extreme values of x and y, putting $f_x = 0$, and $f_y = 0$, we get

$$(a^2 + c^2)x + aby - ap = 0,$$

and $$abx + (b^2 + c^2)y - bp = 0.$$

Solving these two equations, we obtain

$$x = \frac{ap}{a^2 + b^2 + c^2}, \quad y = \frac{bp}{a^2 + b^2 + c^2}.$$

Here $f_{xx} = 2 + \frac{2a^2}{c^2} > 0$, for any values of a and c,

and $$f_{xx} f_{yy} - f_{xy}^2 = 4\left(1 + \frac{a^2}{c^2}\right)\left(1 + \frac{b^2}{c^2}\right) - \frac{4a^2b^2}{c^4} = 4\left(1 + \frac{a^2}{c^2} + \frac{b^2}{c^2}\right) > 0.$$

Hence, the function f has minimum value at the points

$$x = \frac{ap}{a^2 + b^2 + c^2}, \quad y = \frac{bp}{a^2 + b^2 + c^2}.$$

Putting these to (2), we get $$z = \frac{1}{c}\left(p - \frac{a^2 p}{a^2 + b^2 + c^2} - \frac{b^2 p}{a^2 + b^2 + c^2}\right) = \frac{c\,p}{a^2 + b^2 + c^2}.$$

Putting the values of x, y and z in (1), we get

$$f(x, y, z) = \frac{a^2 p^2}{(a^2+b^2+c^2)^2} + \frac{b^2 p^2}{(a^2+b^2+c^2)^2} + \frac{c^2 p^2}{(a^2+b^2+c^2)^2}$$

$$= \frac{a^2p^2+b^2p^2+c^2p^2}{(a^2+b^2+c^2)^2} = \frac{p^2(a^2+b^2+c^2)}{(a^2+b^2+c^2)^2} = \frac{p^2}{a^2+b^2+c^2}.$$

Minimum value of the function f (x, y, z) is

$$\text{Min. value} = \frac{a^2p^2}{(a^2+b^2+c^2)^2} + \frac{b^2p^2}{(a^2+b^2+c^2)^2} + \frac{c^2p^2}{(a^2+b^2+c^2)^2}$$

$$= \frac{a^2p^2+b^2p^2+c^2p^2}{(a^2+b^2+c^2)^2} = \frac{(a^2+b^2+c^2)\,p^2}{(a^2+b^2+c^2)^2} = \frac{p^2}{a^2+b^2+c^2}.$$

3. If the sum of the dimension of a rectangular swimming pool is given, prove that the amount of water in the pool is maximum when it is a cube.

Solution: Let the dimensions of the rectangular swimming pool be l, b, h, and the amount of water in the pool is given by

$$V = lbh, \qquad \text{... (1)}$$

whose sum is given by $l + b + h = \lambda$ (say),

$$\therefore \quad h = \lambda - l - b, \qquad \text{...(2)}$$

With the help of (2) in (1), we get

$$V = lb(\lambda - l - b) = \lambda lb - bl^2 - lb^2$$

Differentiating partially with respect to l and b, respectively, we get

$$\frac{\partial V}{\partial l} = \lambda b - 2lb - b^2 \quad \text{and} \quad \frac{\partial V}{\partial b} = \lambda l - l^2 - 2l\,b.$$

Second order partial derivatives are

$$\frac{\partial^2 V}{\partial l^2} = -2b, \quad \frac{\partial^2 V}{\partial b^2} = -2l, \quad \frac{\partial^2 V}{\partial b\partial l} = \lambda - 2l - 2b \text{ and } \frac{\partial^2 V}{\partial l\partial b} = \lambda - 2l - 2b.$$

For maximum amount of water, putting $\partial V/\partial l = 0$ and $\partial V/\partial l = 0$ give

$$\lambda b - 2lb - b^2 = 0 \quad \text{and} \quad \lambda l - l^2 - 2l\,b = 0.$$

Solving these two equations, we get $l = \lambda/3$ and $b = \lambda/3$. Putting the values of l and b in (2), we get

$$h = \lambda - \frac{\lambda}{3} - \frac{\lambda}{3} = \frac{\lambda}{3}.$$

For maximum amount of water, sufficient conditions to be satisfied are

$$\frac{\partial^2 V}{\partial l^2} < 0 \text{ and } \frac{\partial^2 V}{\partial l^2}\frac{\partial^2 V}{\partial b^2} - \left(\frac{\partial^2 V}{\partial b\partial l}\right)^2 > 0.$$

These conditions are clearly satisfied since l, b, h are positive (being lengths). Hence, the amount of water in the swimming pool is maximum when it is a cube, that is, when

$$l = b = h = \frac{\lambda}{3} \text{ cm}.$$

Maximum amount of water in the pool is $V = \frac{\lambda}{3} \times \frac{\lambda}{3} \times \frac{\lambda}{3} = \frac{\lambda^3}{27} \text{ cm}^3$.

4. Find the extreme value of $x^2 + y^2 + z^2$ connected by the relations $x + z = 1$ and $2y + z = 2$

Solution: A function F of three variables x, y and z, is given by

$$F = x^2 + y^2 + z^2 \quad \text{...(1)}$$

connected by the relations

$$x + z = 1, \quad \text{...(2)}$$

$$2y + z = 2. \quad \text{...(3)}$$

From (2) and (3), we get $z = 1 - x$, and $y = (1 + x)/2$. The function f of (1) reduces to

$$F = x^2 + \frac{1}{4}(1 + x)^2 + (1 - x)^2.$$

Differentiating it with respect to x, we get

$$\frac{dF}{dx} = 2x + \frac{1}{2}(1 + x) - 2(1 - x) = \frac{9x}{2} - \frac{3}{2}.$$

Differentiating it with respect to x, we get

$$\frac{d^2F}{dx^2} = \frac{9}{2} > 0.$$

For the extreme values, putting $dF/dx = 0$, we get

$$\frac{9x}{2} - \frac{3}{2} = 0.$$

Therefore, $x = \frac{1}{3}$. Then from (2) and (3), we get $z = \frac{2}{3}$, and $y = \frac{2}{3}$.

Therefore the stationary point is (1/3, 2/3, 2/3).

$$\text{Min .Value} = \left(\frac{1}{3}\right)^2 + \left(\frac{2}{3}\right)^2 + \left(\frac{2}{3}\right)^2 = \frac{1}{9} + \frac{4}{9} + \frac{4}{9} = 1.$$

1.2.3 Lagrange's Multiplier

When the above method becomes impracticable, Lagrange's method proves very convenient. To find the stationary points, we use Lagrange's method of undetermined multiplier usually denoted by λ. This method is explained as follows:

1. *To find the stationary point of a function f(x, y) under the condition* $\phi(x, y) = 0$

A function f(x, y) of two variables x, y which are connected by the relation

$$\phi(x, y) = 0. \quad \text{....(1)}$$

For the function f(x, y) to have stationary values, it is necessary that

$$\frac{\partial f}{\partial x} = 0 \text{ and } \frac{\partial f}{\partial y} = 0,$$

$$\therefore \quad df = \frac{\partial f}{\partial x}\, dx + \frac{\partial f}{\partial y}\, dy = 0. \qquad ....(2)$$

Also differentiating (1), we get

$$d\phi = \frac{\partial \phi}{\partial x}\, dx + \frac{\partial \phi}{\partial y}\, dy = 0 \qquad ...(3)$$

Multiplying (3) by a parameter λ and adding to (2), we have

$$\left(\frac{\partial f}{\partial x} + \lambda \frac{\partial \phi}{\partial x}\right) dx + \left(\frac{\partial f}{\partial y} + \lambda \frac{\partial \phi}{\partial y}\right) dy = 0$$

This equation will be satisfied if

$$\frac{\partial f}{\partial x} + \lambda \frac{\partial \phi}{\partial x} = 0, \quad \text{and} \quad \frac{\partial f}{\partial y} + \lambda \frac{\partial \phi}{\partial y} = 0.$$

These equations together with $\phi(x, y) = 0$ will determine the values of x, y and λ for which f is stationary.

The function f(x, y) has *maximum* value at the stationary point if

$$\begin{vmatrix} 0 & \phi_x & \phi_y \\ \phi_x & f_{xx} & f_{xy} \\ \phi_y & f_{yx} & f_{yy} \end{vmatrix} > 0,$$

and *minimum* value if

$$\begin{vmatrix} 0 & \phi_x & \phi_y \\ \phi_x & f_{xx} & f_{xy} \\ \phi_y & f_{yx} & f_{yy} \end{vmatrix} < 0.$$

2. *To find the stationary point of a function f(x, y, z) under the condition $\phi(x, y, z) = 0$*

A function f(x, y, z) of three variables x, y and z which are connected by the relation

$$\phi(x, y, z) = 0 \qquad ....(1)$$

For the function f(x, y, z) to have stationary values, it is necessary that

$$\frac{\partial f}{\partial x} = 0, \quad \frac{\partial f}{\partial y} = 0, \quad \frac{\partial f}{\partial z} = 0,$$

$$\therefore \quad \frac{\partial f}{\partial x}\, dx + \frac{\partial f}{\partial y}\, dy + \frac{\partial f}{\partial z}\, dz = 0. \qquad ....(2)$$

Also differentiating (1), we get $d\phi = 0$,

$$\text{or} \quad \frac{\partial \phi}{\partial x}\, dx + \frac{\partial \phi}{\partial y}\, dy + \frac{\partial \phi}{\partial z}\, dz = 0 \qquad .....(3)$$

Multiply (3) by a parameter λ and adding to (2), then

$$\left(\frac{\partial f}{\partial x} + \lambda \frac{\partial \phi}{\partial x}\right) dx + \left(\frac{\partial f}{\partial y} + \lambda \frac{\partial \phi}{\partial y}\right) dy + \left(\frac{\partial f}{\partial z} + \lambda \frac{\partial \phi}{\partial z}\right) dz = 0$$

This equation will be satisfied if

$$\frac{\partial f}{\partial x} + \lambda \frac{\partial \phi}{\partial x} = 0, \quad \frac{\partial f}{\partial y} + \lambda \frac{\partial \phi}{\partial y} = 0, \quad \frac{\partial f}{\partial z} + \lambda \frac{\partial \phi}{\partial z} = 0.$$

These three equations together with (1) will determine the values of x, y, z and λ for which f is stationary.

Working rule

(i) Writing $F = f(x, y, z) + \lambda\, \phi(x, y, z)$, (ii) and obtain the equations as

$$\frac{\partial F}{\partial x} = 0, \quad \frac{\partial F}{\partial y} = 0, \quad \frac{\partial F}{\partial z} = 0,$$

(iii) Solve these equations together with $\phi(x, y, z) = 0$.

The value of x, y, z so obtained will give the stationary value of f(x, y, z). The function f(x, y, z) has *maximum value* at the stationary point if

$$\begin{vmatrix} 0 & \phi_x & \phi_y \\ \phi_x & f_{xx} & f_{xy} \\ \phi_y & f_{yx} & f_{yy} \end{vmatrix} > 0, \quad \begin{vmatrix} 0 & \phi_x & \phi_y & \phi_z \\ \phi_x & f_{xx} & f_{xy} & f_{xz} \\ \phi_y & f_{yx} & f_{yy} & f_{yz} \\ \phi_z & f_{zx} & f_{zy} & f_{zz} \end{vmatrix} < 0,$$

and *minimum value* at the stationary points

$$\text{if} \quad \begin{vmatrix} 0 & \phi_x & \phi_y \\ \phi_x & f_{xx} & f_{xy} \\ \phi_y & f_{yx} & f_{yy} \end{vmatrix} < 0, \quad \begin{vmatrix} 0 & \phi_x & \phi_y & \phi_z \\ \phi_x & f_{xx} & f_{xy} & f_{xz} \\ \phi_y & f_{yx} & f_{yy} & f_{yz} \\ \phi_z & f_{zx} & f_{zy} & f_{zz} \end{vmatrix} < 0.$$

3. *To find stationary point of a function f(x, y, z) under two conditions $\phi_1(x, y, z) = 0$, $\phi_2(x, y, z) = 0$*

A function f(x, y, z) of three variables x, y, z which are connected by the relations

$$\phi_1(x, y, z) = 0, \qquad \ldots(1)$$

and $$\phi_2(x, y, z) = 0. \qquad \ldots(2)$$

For the function f(x, y, z) to have stationary values, it is necessary that

$$\frac{\partial f}{\partial x} = 0, \qquad \frac{\partial f}{\partial y} = 0, \qquad \frac{\partial f}{\partial z} = 0,$$

$$\therefore \qquad df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy + \frac{\partial f}{\partial z} dz = 0. \qquad \ldots(3)$$

Also differentiating (1) and (2) respectively, we get

$$d\phi_1 = \frac{\partial \phi_1}{\partial x}\, dx + \frac{\partial \phi_1}{\partial y}\, dy + \frac{\partial \phi_1}{\partial z}\, dz = 0, \qquad ...(4)$$

$$d\phi_2 = \frac{\partial \phi_2}{\partial x}\, dx + \frac{\partial \phi_2}{\partial y}\, dy + \frac{\partial \phi_2}{\partial z}\, dz = 0. \qquad ...(5)$$

Multiplying (4) by a parameter λ_1 and (5) by another parameter λ_2 and add to (3), then

$$\left(\frac{\partial f}{\partial x} + \lambda_1 \frac{\partial \phi_1}{\partial x} + \lambda_2 \frac{\partial \phi_2}{\partial x}\right) dx + \left(\frac{\partial f}{\partial y} + \lambda_1 \frac{\partial \phi_1}{\partial y} + \lambda_2 \frac{\partial \phi_2}{\partial y}\right) dy + \left(\frac{\partial f}{\partial z} + \lambda_1 \frac{\partial \phi_1}{\partial z} + \lambda_2 \frac{\partial \phi_2}{\partial z}\right) dz = 0$$

This equation will be satisfied if

$$\frac{\partial f}{\partial x} + \lambda_1 \frac{\partial \phi_1}{\partial x} + \lambda_2 \frac{\partial \phi_2}{\partial x} = 0$$

$$\frac{\partial f}{\partial y} + \lambda_1 \frac{\partial \phi_1}{\partial y} + \lambda_2 \frac{\partial \phi_2}{\partial y} = 0, \quad \frac{\partial f}{\partial z} + \lambda_1 \frac{\partial \phi_1}{\partial z} + \lambda_2 \frac{\partial \phi_2}{\partial z} = 0.$$

These three equations together with (1) and (2), $\phi(x, y, z) = 0$ will determine the values of x, y, z and λ for which f is stationary.

Working rule

(i) Writing $F = f(x, y, z) + \lambda_1\, \phi_1(x, y, z) + \lambda_2\, \phi_2(x, y, z)$,

(ii) Obtain the equations $\frac{\partial F}{\partial x} = 0$, $\frac{\partial F}{\partial y} = 0$, $\frac{\partial F}{\partial z} = 0$,

(iii) Solve these equations together with

$\phi_1(x, y, z) = 0$, $\phi_2(x, y, z) = 0$.

The value of x, y, z so obtained will give the stationary value of f(x, y, z).

Worked out Examples

1. Find the minimum value of $x^2 + y^2$ under the condition ax + by = c using Lagrange's multiplier.

Solution: A function f of two variables x, and y, is given by

$$f = x^2 + y^2, \qquad ...(1)$$

under the condition

$$\phi = ax + by - c = 0. \qquad ...(2)$$

Using Lagrange's multiplier λ, we write $F = f + \lambda\, \phi$ gives

$$F = x^2 + y^2 + \lambda\,(ax + by - c). \qquad ...(3)$$

Partially differentiating (3) with respect to x and y respectively, we get

$$F_x = 2x + a\lambda, \qquad F_y = 2y + b\lambda.$$

For the stationary points, putting $F_x = 0$, $F_y = 0$, we get

$$2x + a\lambda = 0, \qquad 2y + b\lambda = 0.$$

Solving these two equations, we get x = (a / b) y. Substituting the values of x in (2), we get

$$y = \frac{bc}{a^2 + b^2}, \qquad x = \frac{ac}{a^2 + b^2}, \text{ and } \lambda = -\frac{2c}{a^2 + b^2}.$$

Therefore, the stationary point is $\left(\frac{ac}{a^2 + b^2}, \frac{bc}{a^2 + b^2}\right)$. The minimum value of $x^2 + y^2$ is

$$\text{Min. value} = \frac{a^2 c^2}{(a^2 + b^2)^2} + \frac{b^2 c^2}{(a^2 + b^2)^2} = \frac{c^2}{a^2 + b^2}.$$

2. Find the minimum value of $x^2 + xy + y^2 + 3z^2$ under the condition $x + 2y + 4z = 60$ using Lagrange's multiplier.

Solution: A function f of three variables x, y and z, is given by

$$f = x^2 + xy + y^2 + 3z^2, \qquad \text{...(1)}$$

under the condition

$$\phi = x + 2y + 4z - 60 = 0. \qquad \text{...(2)}$$

Using Lagrange's multiplier λ, $F = f + \lambda\,\phi$ gives

$$F = x^2 + xy + y^2 + 3z^2 + \lambda\,(x + 2y + 4z - 60). \qquad \text{...(3)}$$

Partially differentiating (3) with respect to x, y and z respectively, we get

$$F_x = 2x + y + \lambda, \qquad F_y = x + 2y + 2\lambda, \qquad F_z = 6z + 4\lambda$$

For the stationary points, putting $F_x = 0$, $F_y = 0$, $F_z = 0$, we get

$$2x + y + \lambda = 0, \qquad x + 2y + 2\lambda = 0, \quad \text{and} \quad 6z + 4\lambda = 0.$$

Solving these three equations and (2), we get

$$\lambda = \frac{-90}{7}, \quad x = 0, \quad y = \frac{90}{7}, \qquad z = \frac{60}{7}.$$

Therefore the stationary point is $\left(0, \frac{90}{7}, \frac{60}{7}\right)$. From (1) and (2), we get

$$f_x = 2x + y, \qquad f_y = x + 2y, \quad f_z = 6z$$

$$f_{xx} = 2, \qquad f_{xy} = 1, \qquad f_{xz} = 0, \quad f_{yx} = 1, f_{yz} = 0,$$

$$f_{yy} = 2, \qquad f_{zx} = 0, \qquad f_{zy} = 0, \quad \phi_x = 1, \ \phi_y = 2, \ \phi_z = 4.$$

Now

$$\begin{vmatrix} 0 & \phi_x & \phi_y \\ \phi_x & f_{xx} & f_{xy} \\ \phi_y & f_{yx} & f_{yy} \end{vmatrix} = \begin{vmatrix} 0 & 1 & 2 \\ 1 & 2 & 1 \\ 2 & 1 & 2 \end{vmatrix} = -8 < 0,$$

and

$$\begin{vmatrix} 0 & \phi_x & \phi_y & \phi_z \\ \phi_x & f_{xx} & f_{xy} & f_{xz} \\ \phi_y & f_{yx} & f_{yy} & f_{yz} \\ \phi_z & f_{zx} & f_{zy} & f_{zz} \end{vmatrix} = \begin{vmatrix} 0 & 1 & 2 & 4 \\ 1 & 2 & 1 & 0 \\ 2 & 1 & 2 & 0 \\ 4 & 0 & 0 & 6 \end{vmatrix} = -12 < 0.$$

Therefore, f has minimum value at (0, 90/7, 60/7),

$$\text{Min. value} = \frac{8100}{49} + \frac{10800}{49} = \frac{2700}{7}.$$

3. Find the minimum value of $x^2 + y^2 + z^2$ subject to the condition $\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = 1$ using Lagrange's multiplier.

Solution: A function f of three variables x, y and z, is given by

$$f = x^2 + y^2 + z^2, \qquad \ldots(1)$$

subject to the relation

$$\phi = \frac{1}{x} + \frac{1}{y} + \frac{1}{z} - 1 = 0. \qquad \ldots(2)$$

Then, we write

$$F = x^2 + y^2 + z^2 + \lambda\left(\frac{1}{x} + \frac{1}{y} + \frac{1}{z} - 1\right), \qquad \ldots(3)$$

where λ is a Lagrange's multiplier.

Partially differentiating (3) with respect to *x*, *y* and z respectively, we get

$$F_x = 2x - \frac{\lambda}{x^2}, \quad F_y = 2y - \frac{\lambda}{y^2}, \quad F_z = 2z - \frac{\lambda}{z^2}.$$

For the stationary points, putting $F_x = 0$, $F_y = 0$ and $F_z = 0$, we get

$$2x - \frac{\lambda}{x^2} = 0, \quad 2y - \frac{\lambda}{y^2} = 0 \text{ and } \quad 2z - \frac{\lambda}{z^2} = 0.$$

Solving these equations and (2), we find

$$x = \left(\frac{\lambda}{2}\right)^{\frac{1}{3}}, y = \left(\frac{\lambda}{2}\right)^{\frac{1}{3}} \text{ and } z = \left(\frac{\lambda}{2}\right)^{\frac{1}{3}}.$$

Substituting the values of x, y and z in (2), we get

$$\left(\frac{2}{\lambda}\right)^{\frac{1}{3}} + \left(\frac{2}{\lambda}\right)^{\frac{1}{3}} + \left(\frac{2}{\lambda}\right)^{\frac{1}{3}} - 1 = 0$$

or

$$3\left(\frac{2}{\lambda}\right)^{\frac{1}{3}} = 1,$$

$\therefore$

$$\lambda = 54.$$

Then $x = 3$, $y = 3$ and $z = 3$. Thus the stationary point is (3, 3, 3).

Partially differentiating (1) and (2) with respect to x, y and z respectively, we get

$$f_x = 2x, \quad f_y = 2y, \quad f_z = 2z,$$

$$\phi_x = -\frac{1}{x^2} = -\frac{1}{9}, \quad \phi_y = -\frac{1}{y^2} = -\frac{1}{9}, \quad \phi_z = -\frac{1}{z^2} = -\frac{1}{9},$$

$$f_{xx} = 2, \quad f_{xy} = 0, \quad f_{xz} = 0,$$

$$f_{yx} = 0, \quad f_{yy} = 2, \quad f_{yz} = 0,$$

$$f_{zx} = 0, \quad f_{zy} = 0, \quad f_{zz} = 2$$

Now

$$\begin{vmatrix} 0 & \phi_x & \phi_y \\ \phi_x & f_{xx} & f_{xy} \\ \phi_y & f_{yx} & f_{yy} \end{vmatrix} = \begin{vmatrix} 0 & -1/9 & -1/9 \\ -1/9 & 2 & 0 \\ -1/9 & 0 & 2 \end{vmatrix}$$

$$= \frac{1}{9}\begin{vmatrix} -1/9 & 0 \\ -1/9 & 2 \end{vmatrix} - \frac{1}{9}\begin{vmatrix} -1/9 & 2 \\ -1/9 & 0 \end{vmatrix}$$

$$= \frac{1}{9}\left(-\frac{2}{9}\right) - \frac{1}{9}\left(\frac{2}{9}\right) = -\frac{4}{81} < 0.$$

$$\begin{vmatrix} 0 & \phi_x & \phi_y & \phi_z \\ \phi_x & f_{xx} & f_{xy} & f_{xz} \\ \phi_y & f_{yx} & f_{yy} & f_{yz} \\ \phi_z & f_{zx} & f_{zy} & f_{zz} \end{vmatrix} = \begin{vmatrix} 0 & -1/9 & -1/9 & -1/9 \\ -1/9 & 2 & 0 & 0 \\ -1/9 & 0 & 2 & 0 \\ -1/9 & 0 & 0 & 2 \end{vmatrix}$$

$$= \frac{1}{9}\begin{vmatrix} -1/9 & -1/9 & -1/9 \\ -1/9 & 0 & 2 \\ 0 & 0 & 2 \end{vmatrix} + 2\begin{vmatrix} 0 & -1/9 & -1/9 \\ -1/9 & 2 & 0 \\ -1/9 & 0 & 2 \end{vmatrix}$$

$$= \frac{1}{9}\left(-\frac{1}{9}\right)\left(-\frac{2}{9}\right) + 2\left[\left(\frac{1}{9}\right)\left(-\frac{2}{9}\right) - \frac{1}{9}\left(\frac{2}{9}\right)\right]$$

$$= \frac{2}{729} - \frac{8}{81} = \frac{2-72}{729} = -\frac{70}{729} < 0.$$

Therefore, f has minimum value at (3, 3, 3), Min. value = 9 + 9 + 9 = 27.

1.2.4 Applications in Optimization of Function of Two Variables with One Constraint

Sometimes it is required to find the stationary values of a function of two and three independent variables connected by some given conditions. With the help of given conditions, the function can reduce to a function of a single variable *x* or two variables *x* and *y*.

Stationary points of the function f(x, y) under the condition $\phi(x, y) = 0$

A function F of two variables x and y is given by

$$F = f(x, y), \quad ...(1)$$

under the condition $\quad \phi(x, y) = 0. \quad ...(2)$

The function F of (1) is reduced to a function of a single variable x with the help of (2). To obtain stationary points, putting $\frac{dF(x)}{dx} = 0$,

which gives the value of x. After substituting the value of x in (2), we get y and thus will be obtain the stationary points.

Worked out Examples

1. Find the two numbers such that the sum of whose squares is minimal and whose sum is 156.

Solution: Let the two numbers be x and y such that $x + y = 156$. Thus, we can express y in terms of x as:

$$y = 156 - x. \quad ... (1)$$

We are given that the sum of x^2 and y^2 is to be minimized. Therefore, we have

$$S = x^2 + y^2. \quad ....(2)$$

With the use of (1) in (2), we write

$$S = x^2 + (156 - x)^2 \quad ....(3)$$

Differentiating it with respect to x, we obtain

$$\frac{dS}{dx} = 2x - 2(156 - x).$$

Further differentiation, we get

$\frac{d^2S}{dx^2} = 2 + 2 = 4 > 0$. This indicates that S has a minimum value.

For the values of x, putting $dS/dx = 0$ gives

$$2x - 2(156 - x) = 0$$

or $\quad 4x = 312, \quad \therefore x = 78$

Substituting $x = 78$ in (1), we get $y = 78$. Hence, the required numbers are 78 and 78.

2. A farmer with 400 meter of fencing wants to enclose a rectangular plot that boarder on a straight river. If the farmer doesnot fence the side along the river, what is the largest area that can be enclosed?

Solution: Let the x and y be length and breadth of the rectangular plot, and its area

$$A = xy, \quad ... (1)$$

and the farmer wants to enclose the area of the perimeter

$$x + y + x = 400,$$

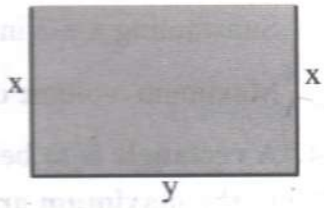
or $$2x + y = 400,$$

$$\therefore \quad y = 400 - 2x \qquad \text{... (2)}$$

With use of (2) in (1), we gete

$$A = x(400 - 2x) = 400x - 2x^2 \qquad \text{....(3)}$$

Differentiating it with respect to x, we obtain



$$\frac{dA}{dx} = 400 - 4x.$$

Further differentiation, we get

$$\frac{d^2A}{dx^2} = -4 < 0.$$ This indicates that A has a maximum value.

For the values of x, putting $dA/dx = 0$ gives $400 - 4x = 0$,

or $4x = 400$, $\therefore x = 100$.

Substituting $x = 100$ in (2), we get $y = 400 - 2 \times 100 = 200$.

Maximum area $A = 100 \times 200 = 20000 \text{ m}^2$.

3. An open box is to be made by cutting small congruent squares from the corners of 16 by 16 chart paper and bending up the sides. Find the dimensions of the box that the maximize the volume.

Solution: Let the x be height and y be the base of the box, and its volume is

$$V = xy^2, \qquad \text{... (1)}$$

as shown in the figure, such that $x + y + x = 18$,

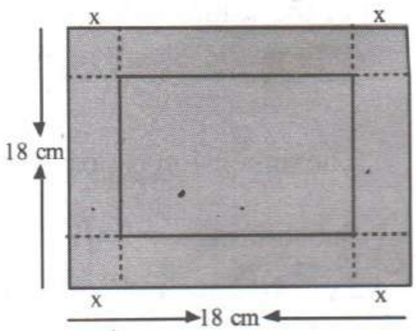
or $$2x + y = 18,$$

$$\therefore \quad y = 18 - 2x. \qquad \text{... (2)}$$

With use of (2) in (1), we gete

$$V = x(18 - 2x)^2 = 324x - 72x^2 + 4x^3 \qquad \text{....(3)}$$

Differentiating it with respect to x, we obtain



$$\frac{dV}{dx} = 324 - 144x + 12x^2.$$

Further differentiation, we get

$$\frac{d^2V}{dx^2} = -144 + 24x.$$

For the values of x, putting $dS/dx = 0$ gives

$$324 - 144x + 12x^2 = 0,$$

or $\quad x^2 - 12x + 27 = 0$, or $(x - 3)(x - 9)$ $\quad \therefore\ x = 3$ and 9.

When $x = 9$, $\dfrac{d^2V}{dx^2} = -144 + 24 \times 9 = 72 > 0$. This indicates that V has minimum value.

When $x = 3$, $\dfrac{d^2V}{dx^2} = -144 + 24 \times 3 = -72 < 0$. This indicates that V has maximum value.

Substituting $x = 3$ in (2), we get $y = 18 - 2 \times 3 = 12$.

Maximum volume of the box is $V = 3 \times 12 = 36\ \text{cm}^3$.

4. A rectangle is to be inscribed a semicircle of radius 4. Find the dimensions of the rectangle and the maximum area.

Solution: Let the y be the height and (2x) be the length of the rectangle. The area A of the rectangle is given by

$$A = (2x)\, y \qquad \ldots (1)$$

Since the rectangle is inscribed in the semicircle,its vertices satisfies the semi circle

$$x^2 + y^2 = 16,$$

$$\therefore \quad y = \sqrt{16 - x^2}. \qquad \ldots (2)$$

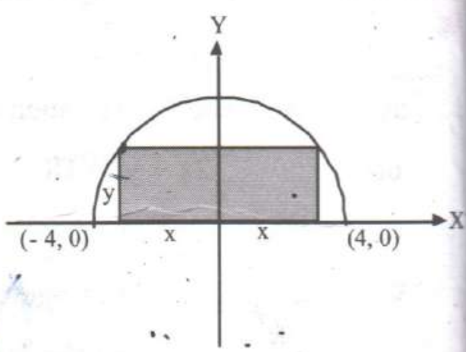
Substituting equation (2) in equation (1), we get

$$A = 2x\sqrt{16 - x^2}. \qquad \ldots (3)$$

Differentiating it with respect to x, we obtain

$$\frac{dA}{dx} = -\frac{2x^2}{\sqrt{16 - x^2}} + 2\sqrt{16 - x^2}$$

$$= \frac{32 - 4x^2}{\sqrt{16 - x^2}}$$



Further differentiation, we get

$$\frac{d^2A}{dx^2} = \frac{64x - 2x^3}{(16 - x^2)^{3/2}} - \frac{2x}{\sqrt{16 - x^2}}.$$

For the values of x, putting $dA/dx = 0$ gives $32 - 4x^2 = 0$,

or $\quad x^2 = 8$, $\quad \therefore\ x = \pm 2\sqrt{2}$.

The dimension is positive, $x = 2\sqrt{2}$, $\dfrac{d^2A}{dx^2} < 0$. This indicates that A has a maximum value.

Substituting $x = 2\sqrt{2}$ in (2), we get $y = \sqrt{16 - 8} = 2\sqrt{2}$. The dimensions of the rectangle are $2x = 4\sqrt{2}$ cm and $2\sqrt{2}$ cm, and Maximum area $A = 4\sqrt{2} \times 2\sqrt{2} = 16\ \text{cm}^2$.

Exercise - 3

1. Examine and find the maximum and minimum values of

   (i) $x^3 + y^3 - 3axy$ (ii) $x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$

   (iii) $xy(a - x - y)$ (iv) $2(x - y)^2 - x^4 - y^4$

   (v) $4x^2 - xy + 4y^2 + x^3y + xy^3 - 4$ (vi) $16 - (x + 2)^2 - (y - 2)^2$

2. Examine and find the maximum and minimum values of

   (i) $y^2 + 2z^2 - 5x^4 + 4x^5$ (ii) $-3x^2 + 6xz + 4y - 2y^2 - 6z^2$

   (iii) $35 - (2x + 3)^2 - (y - 4)^2 - (z + 1)^2$ (iv) $8z + 2x^2 + 3y^2 + 4z^2 - 3xy$

3. Show that following functions have a minimum at the points indicated

   (i) $x^2 + y^2 + z^2 + 2xyz$ at $(0, 0, 0)$ (ii) $x^4 + y^4 + z^4 - 4xyz$ at $(1, 1, 1)$

4. Find the extreme values for the function $x^2 + y^2$ under the condition $x + 4y = 2$

5. Find the extreme values of $48 - (x - 5)^2 - 3(y - 4)^2$ subject to the condition $x + 3y = 9$

6. Find the extreme values of $x^2 + y^2 + z^2$ subject to the condition $x + y + z = 1$

7. Find the maximum and minimum values of xy^2 under the condition $x + y = 1$

8. Determine the maximum value of xyz under the condition that $x + y + z = 8$ using Lagrange's multiplier

9. Determine the minimum value of $x^2 + y^2 + z^2$ subject to the conditions $x + y + z - 1 = 0$ and $xyz + 1 = 0$ using Lagrange's multiplier

10. Find the minimum value $x^2 + y^2 + z^2$ when $x + y + z = 3a$ using Lagrange's multiplier

11. If the sum of three positive numbers is cube of 2, what is the maximum value of their product?

12. Find the maximum area of a rectangular piece of land that can be surrounded by a fence that is 12 meters long.

13. A rectangular-shaped playground ends in two semicircles. Given that the perimeter of the playground is 594 m, determine its maximum area.

14. Find a point within a triangle such that the sum of the squares of its distances from the three angular points (2, 3), (3, 5) and (10, 7) is minimum.

Answers

1. (i) Max. at (a, a); Max. value = $-a^3$

   (ii) Max. at (4, 0), Max. value = 112 ; Min. at (6, 0), Min. value = 108

   (iii) Extreme point $\left(\frac{a}{3}, \frac{a}{3}\right)$; Extreme value = $\frac{a^3}{27}$, Max. value for a > 0 and Min. value for a < 0

   (iv) Max. at $(\sqrt{2}, -\sqrt{2})$ and at $(-\sqrt{2}, \sqrt{2})$; Max. value = 8

   (v) Max. at $\left(\frac{3}{2}, \frac{-3}{2}\right)$ and at $\left(-\frac{3}{2}, \frac{3}{2}\right)$ Max.. value = $\frac{49}{8}$, Min. at (0, 0); Min. value = - 4

   (vi) Max. at (- 2, 2); Max. value = 16

2. (i) Min. at (1, 0, 0); Min. value = - 1 (ii) Max. at (0, 1, 0); Max. value = 2

   (iii) Max. at $\left(-\frac{3}{2}, 4, -1\right)$; Max. value = 35 (iv) Min. at (0, 0, - 1); Min. value = 4

4. Min. value = $\frac{4}{27}$ 5. Max. value = 32 6. Min. value = $\frac{1}{3}$

7. Max. value = $\frac{4}{27}$ 8. Max. value = $\frac{512}{27}$, Min. value = 0 9. Min. value = 3

10. Min. value = $3a^2$ 11. $\frac{512}{27}$ 12. $9m^2$ 13. $\frac{88209}{\pi} m^2$ 14. (5, 5)

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